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**EmpowerEd: Think. Learn. Achieve. – An Accessible Online Learning Platform for
Students with Disabilities**

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CHAPTER 1

INTRODUCTION

Background of the Study

Education is a fundamental right that should be accessible to all learners regardless of their physical, sensory, or cognitive conditions. According to UNESCO (2026), inclusive education aims to ensure that all learners are given equal opportunities and that barriers preventing students from fully participating in the educational process are removed. However, students with disabilities continue to face various challenges in accessing quality education, particularly in online and technology-based learning environments.

As educational institutions increasingly adopt digital platforms for teaching and learning, the need for accessible and inclusive systems has become more important. According to Lomellini et al. (2025), many online learning platforms are designed for general users and often do not fully address the specific needs of students with disabilities, which may affect their academic participation and performance.

Students with visual disabilities may encounter difficulties in reading learning materials that lack text-to-speech functionality or adjustable font sizes. Likewise, students with hearing disabilities may struggle to understand video-based lessons when subtitles or captions are not available. In the study of Lomellini et al. (2025), inaccessible content such as uncaptioned videos, scanned PDF files, and materials incompatible with assistive

technologies were identified as major barriers to online learning for students with disabilities.

The integration of technology in education has provided opportunities for more flexible and self-paced learning. According to UNESCO's guidelines on inclusive learning, digital platforms should incorporate assistive tools and accessible features to support learners with diverse needs and promote equal educational opportunities. Therefore, educational platforms must include features that help reduce learning barriers and improve academic success among students with disabilities.

In response to this concern, the proposed study entitled "EmpowerEd: Think. Learn. Achieve. Developing an Accessible Online Learning Platform for Students with Disabilities" aims to develop a system specifically designed to support learners with disabilities. The platform includes accessibility features such as text-to-speech, subtitles, downloadable learning materials, and customizable font sizes. In addition, it provides tools for teachers to upload lessons, assign tasks, and monitor student progress effectively.

This proposed study also supports Sustainable Development Goal 4 (SDG 4): Quality Education, which aims to "ensure inclusive and equitable quality education and promote lifelong learning opportunities for all." A key target of SDG 4 emphasizes equal access to education for vulnerable groups, including persons with disabilities, and the development of inclusive and accessible learning environments. By creating an online learning platform specifically designed for students with disabilities, the study contributes



to the achievement of SDG 4 through the promotion of accessible digital education and equal learning opportunities for all learners.

This study is significant because it seeks to contribute to the promotion of inclusive and accessible education through the use of technology. By comparing the academic performance of students who use the EmpowerEd platform with those who continue using traditional learning methods, the researchers aim to determine the effectiveness of the proposed system in improving the learning experience and academic outcomes of students with disabilities.

Rationale of the Study

The persistent digital divide remains a significant barrier to the realization of inclusive education, particularly for students with physical, sensory, or cognitive impairments. While global frameworks like the UN Convention on the Rights of Persons with Disabilities mandate equal access, current legal structures often fail to provide necessary individualized accommodations. In the Philippines, despite the existence of Republic Act 10754, students with disabilities continue to encounter systemic obstacles and persistent stigma that hinder full participation.

Existing Learning Management Systems (LMS) are frequently developed with a general user base in mind, resulting in poor interface design and a lack of built-in assistive features. Research by Fithriyaningrum et al. (2022) highlights that many academic platforms contain widespread barriers such as missing alternative text, inadequate color contrast, and poor keyboard navigation, which transform a digital classroom into an



inaccessible space. These technical issues lead to low student engagement, underscoring the urgent need for adaptable applications that fit the specific technology setup of the Philippine education system.

The "EmpowerEd" platform is developed to address these gaps by integrating the principles of Universal Design for Learning (UDL) and Smart Learning Environments (SLE). According to Almeqdad et al. (2023), adopting a UDL framework proactively reduces learning barriers by providing multiple means of representation, such as auditory texts and high-contrast visuals, rather than relying on reactive accommodations. Furthermore, by incorporating the nine core quality attributes of smart classrooms—including technology accessibility, system simplicity, and interconnectivity—the platform ensures that educational intelligence is accessible to both teachers and students.

This study is driven by the necessity to move from theory to practice. By developing a specialized web-based system that supports AI-driven personalization and offline accessibility, the researchers aim to provide a scalable solution that ensures students with disabilities are not left behind. Ultimately, this research seeks to justify the integration of advanced assistive technologies as a fundamental requirement for achieving true equity in the digital academic landscape



Statement of the Problem

This study aims to develop “*EmpowerEd: Think. Learn. Achieve.*”, an accessible online learning platform designed to support students with disabilities. Specifically, the study sought to answer the following questions:

1. What is the level of academic performance of students with disabilities before and after the use of *EmpowerEd: Think. Learn. Achieve.* in terms of:
 - a. Pre-test scores
 - b. Post-tests scores
2. What is the level of usability, accessibility, and effectiveness of the learning platform as perceived by students and teachers in terms of:
 - a. Navigation and Usability
 - b. Content accessibility
 - c. User satisfaction
3. What is the significant difference between using “*EmpowerEd: Think. Learn. Achieve.*” and traditional online learning platforms in terms of:
 - a. Academic Performance
 - b. Student engagement
 - c. Learning efficiency



Hypothesis

The objective of this study is to examine the effectiveness of the EmpowerEd accessible learning platform in improving the learning performance of students with disabilities. The hypotheses will be formulated and tested at a significance level of 0.05.

1. The EmpowerEd accessible learning platform has no significant effect on the learning performance of students with disabilities. (H_0)

2. The EmpowerEd accessible learning platform has a significant effect on improving the learning performance of students with disabilities. (H_1)

Significance of the Study

The study aims to examine the effectiveness of the EmpowerEd accessible learning platform in improving the learning performance of students with disabilities.

The findings of the study not only provide valuable information but also benefit the individuals as follows:

The Students with Disabilities. The outcome of the study may serve as their basis for utilizing accessible learning tools that promote independence and cater to their specific sensory needs, which will lead to a more equitable and effective educational experience.



The Educators. Through the analysis presented in this study, it will enable them to utilize a streamlined toolset for managing diverse learning requirements, which results in reducing the need for manual adaptation of every lesson and providing them with a more efficient teaching workflow.

The School Administrators. The outcome of the study may serve as their basis for implementing inclusive education policies through technological integration, providing their institutions with a data-driven approach to meeting accessibility standards.

The Parents and Guardians. Through the features of the platform, it will enable them to have a transparent way to monitor their child's academic progress and ensures that learning can continue effectively outside the physical classroom environment.

The Future Developers. The formulation of this study can be used as a technical blueprint or reference for creating accessible UI/UX designs. In addition, developers may utilize this to further enhance inclusive web systems for diverse user groups.

The Future Researchers. The formulation of this study can be used as a fundamental source of knowledge or reference for future researchers. In addition, future researchers may also utilize this to obtain more comprehensive data regarding the long-term effectiveness of assistive digital technologies in education.



Scope and Delimitation

This study aims to evaluate the effectiveness of EmpowerEd, an online learning platform designed to provide accessible and flexible learning tools for students with disabilities. The primary research respondents of this study will be students currently enrolled in Special Education (SPED) programs or inclusive education settings for the academic year 2025-2026. The participants are delimited to 60 students, specifically those with visual, hearing, or learning disabilities (such as Dyslexia or ADHD).

The researchers will utilize a Quasi-Experimental Research Design to conduct the study. The 60 participants will be divided into two groups: an Experimental Group of 30 students who will use the EmpowerEd platform, and a Control Group of 30 students who will continue with traditional learning methods, such as printed modules or standard websites. Data will be collected through standardized assessments and performance monitoring over the course of one academic term. This study will be conducted by researchers pursuing a degree in Information Technology.

The primary technical components of this experimental study include the development of a web-based system featuring Text-to-Speech (TTS), video captioning, adjustable font sizes, and high-contrast UI. The platform also includes administrative tools for teachers to upload lessons, assign tasks, and monitor student progress in real-time. To ensure usability in various environments, the platform will also support downloadable materials for offline use.



The study's weaknesses include the variation in the technical specifications of the students' personal hardware, such as the difference between high-end laptops and budget smartphones, which may affect the platform's performance. Furthermore, while the platform offers offline materials, the stability of the local internet connection remains a factor for initial setup and data syncing. The study may also be limited by the varying severity of individual learning disabilities, which could influence how different students interact with the user interface regardless of the provided accessibility features. Finally, the study will be time-consuming as it requires consistent monitoring of student progress over several weeks to gather sufficient data for comparison.

Theoretical Framework

This study utilizes the Universal Design for Learning (UDL) framework developed by the Center for Applied Special Technology (CAST). The framework asserts that educational environments must be inherently flexible to accommodate individual learning differences. UDL operates on three core principles: providing learners with multiple means of engagement, representation, and action and expression.

In the context of EmpowerEd, UDL serves as the primary blueprint for the system's architecture and UI/UX design. Because students with diverse disabilities process information differently, EmpowerEd applies UDL principles to actively dismantle the structural digital barriers that limit their educational access. This application is evident in the platform's features, which directly correspond to UDL's core principles. To provide multiple means of representation, EmpowerEd integrates text-to-speech modules,



dynamic video subtitles, and customizable font sizes. This ensures that visually and hearing-impaired students can perceive and access the exact same educational content without relying on external workarounds. Additionally, to provide multiple means of engagement, the system offers a simplified, highly accessible navigation interface. This design allows students to access materials repeatedly and learn at their own pace, which directly supports users with learning or cognitive disabilities.

Ultimately, the UDL framework justifies the technical development of EmpowerEd by establishing that the digital application must adapt to the user's specific disability profile, rather than forcing the user to navigate a rigid, inaccessible system.

Conceptual Framework

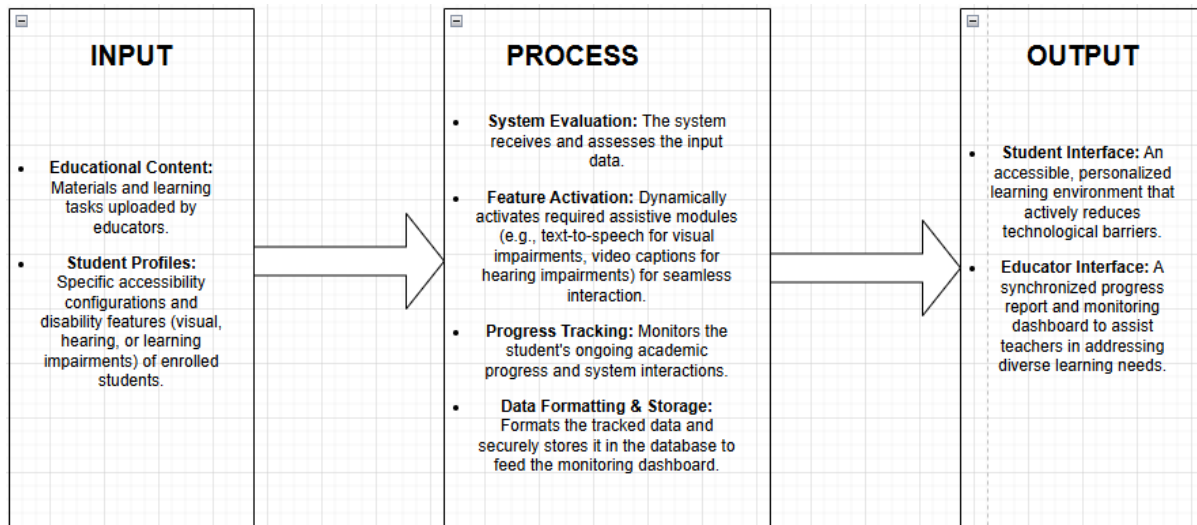


Figure 1: Input-Process-Output Model

The study utilizes the Input-Process-Output (IPO) model to visualize the data flow and the system architecture of EmpowerEd. The input phase consists of the educational materials and tasks uploaded by the educators, the implementation of specific disability features of the enrolled students for visual, hearing, or learning impairments. In the process phase, once the system receives the input, the specific disability features for the student such as text-to-speech modules for visual impairments or enabling video captions for hearing impairments to ensure seamless interaction. This is done as the system tracks the student's academic progress, formats the data, and stores it in the database for the teacher's monitoring dashboard. Finally, the output phase provides an accessible, personalized learning interface that actively reduces technological barriers, alongside a synchronized progress report that assists educators in effectively addressing diverse learning needs.



Definition of Terms

This aims to determine the key or important terms in the study that are clearly defined.

Accessibility. A software evaluation criterion and legal standard measuring how effectively a platform provides equitable access and reduces learning barriers.

Accessible. Systems designed without digital or sensory barriers, allowing individuals with disabilities to independently engage with content.

Automatic Speech Recognition (ASR). Speech technology provides real-time transcription to facilitate hands-free note-taking for students with disabilities.

Hearing Difficulties. Sensory impairments that limit access to audio lessons, requiring visual accommodations like real-time captioning or subtitles.

Inclusive. Ensuring all learners regardless of identity or functional profile have equal opportunities and a safe, supportive digital learning space.

Online Learning Platform. A web-based Learning Management System (LMS) with modular components designed for flexible, technology-enhanced education.

Reading Difficulties. Cognitive or sensory challenges (like Dyslexia) where students struggle to parse standard text without assistive features.

SPED. Specialized or inclusive educational settings designed for learners with visual, hearing, or learning disabilities (e.g., Dyslexia, ADHD).



Students with Disabilities. Learners facing physical, sensory, or cognitive impairments that create barriers to accessing standard digital education. They serve as the primary participants in this study and will mainly use the platform.

Teachers. Educators who utilize the platform to upload lessons, assign tasks, and monitor diverse student progress in real-time without manual adaptations.

Text-to-Speech (TTS). Assistive technology that converts written text into spoken audio to aid students with visual or reading difficulties.

UDL (Universal Design for Learning). A framework that proactively reduces learning barriers from the outset by offering multiple means of representation, action, and expression.

UI/UX. Layout and interaction design focusing on simple layouts, clear text, and high-contrast visuals to improve navigation and readability for diverse users.

Usability. A software quality standard evaluating how intuitively and easily teachers and students with disabilities can navigate and satisfy their needs on the interface.

Visual Impairments. Sensory conditions limiting standard interface viewability, requiring screen readers, large components, or high-contrast display modes.

CHAPTER 2

REVIEW OF THE RELATED LITERATURES AND STUDIES

This chapter provides a review of literature and studies on authors with published works in both local and foreign categories, that are correlated and contributing with the study's chosen topic.

Inclusive Education

Inclusive education is an approach based on equal opportunities and right of every learner to quality education, regardless of his/her personal traits, abilities, disabilities, or background. The study suggests that, in order to ensure learners' accessibility to the educational environment, it is important to create programs that will meet the needs of students who have special educational requirements. This refers to a basic principle in providing an inclusive environment for learners with disabilities, known as the principle of conformity. This principle requires consideration of the different types of disabilities of each student and the design of teaching methods using appropriate pedagogical approaches. (Atajonova and Kasimakhunova 2025)

Additionally, the concept of inclusive education is characterized by other vital principles such as equality, respect for diversity, multi-track instructions, and support. Through these principles, the system addresses the guarantee that all learners with different needs are catered to and provided with quality education. The main goal of inclusive education is to create an educational environment that ensures learners are receiving quality education regardless of their abilities. (Nurali, 2025)



Learners are not the only ones who benefit from inclusive education; teachers, schools, and society at large also benefit. According to Special and Inclusive Education: Perspectives, Challenges, and Prospects (2021), inclusive education ensures equal access to quality education for all learners, encourages collaboration among general and special education teachers, and improves instructional practices through an inclusive approach. It also supports the development of more responsive and flexible school systems while fostering a culture of acceptance and participation. Furthermore, inclusive education contributes to building a more inclusive society by preparing learners to become active participants in society.

The UNESCO also recognizes inclusive education is important and released educational books and resource material about it. One of it is a resource pack titled Reaching Out to All Learners. It emphasizes that schools and educational systems should remove educational barriers by adapting according to the needs of diverse learners. The book highlights the importance of integrating flexible instruction, collaborating among educators, and using accessible educational resources to support students with disabilities. On the other hand, the book by Blicher et. al (2025) also stresses the importance of creating inclusive and equitable learning materials that represent diverse learners and promote accessibility. This confirms that inclusive education is not only about solving the physical barriers but also about ensuring that technologies, learning content, and even teaching strategies are designed in ways to accommodate the different needs and abilities of learners.

Students With Disabilities

Digital and modern academic landscapes require a profound shift toward dismantling the technological and pedagogical barriers encountered by individuals with special needs. At its core, establishing equality relies directly on fostering total inclusiveness within dynamic learning settings (Altınay & Altınay, 2024). Traditional frameworks frequently default to rigid instructional models that treat academic difficulties as personal physical deficits. However, sustainable and contemporary education mandates a transition toward flexible strategies that acknowledge intellectual diversity and establish a rights-based baseline for access (Altınay & Altınay, 2024). To eliminate institutional barriers, modern digital education must be treated not simply as an automated alternative to traditional settings, but as an intentionally constructed mechanism designed to safeguard equitable opportunities across all academic tiers (Altınay & Altınay, 2024).

The integration of advanced digital tools plays an instrumental role in elevating student engagement, particularly for individuals navigating cognitive or physical limitations. Research indicates that when digital platforms incorporate specialized multi-modal assets, they actively counter the isolation and stress often associated with self-paced or remote learning environments. To prevent monotonous content delivery from inducing cognitive fatigue, digital ecosystems can leverage highly interactive mechanisms. Specifically, the introduction of gamification frameworks—such as structured self-development applications—has been proven to significantly maximize cognitive learning outcomes by rendering abstract, complex subject matter into highly

tangible, engaging, and manageable visual modules (Sherafim, 2026). This interactive layer compensates for functional weaknesses, driving student focus and creating scalable pathways toward academic autonomy (Sherafim, 2026).

Beyond gamified elements, the optimization of instructional delivery heavily relies on the technical execution of adaptive learning architectures. In localized settings, the application of differentiated instructional tools demonstrates that providing students with elastic, tailored resources drastically elevate overall reading comprehension and analytical focus (Langcuyan et al., 2024). When learning materials are dynamically adjusted to mirror distinct user profiles, student comprehension and engagement levels consistently reach outstanding performance benchmarks (Langcuyan et al., 2024). This fluidity validates the premise that adapting standard materials—such as providing high-contrast text settings, automated narration features, and customizable layouts—acts as a localized catalyst for narrowing achievement gaps and sustaining learner momentum (Langcuyan et al., 2024).

To achieve lasting educational equality, these standalone features must be unified within a broader framework of digital institutional inclusion, known as e-Inclusion. Developing inclusive online education is a highly complex task that demands the creation of an entirely new pedagogy to address hidden disparities (Slootman et al., 2023). Virtual learning spaces are not inherently inclusive; without meticulous engineering, failing user interfaces and inadequate digital designs can exacerbate existing inequities and completely alienate underserved students (Slootman et al., 2023). By executing an intentional design that maps technological features directly to the pedagogical and

physical needs of the user, educators can foster a profound sense of institutional belonging (Slootman et al., 2023). This guarantees that every student—regardless of their identity, unique body, or functional profile—is provided a safe, supportive digital environment to actively participate, learn, and thrive (Slootman et al., 2023).

Online Learning Platforms

Online learning platforms have become essential tools in modern education because they support flexible, accessible, and technology-enhanced learning environments. According to Pagtaconan, W. C. R. and Mercado, C. A. (2021), Learning Management Systems (LMS) are composed of interconnected components such as content management, communication tools, assessment systems, user roles, and administrative modules. Their study emphasized that LMS platforms should be modular, scalable, and adaptable to institutional needs to ensure effective implementation in educational institutions. They also explained that the success of virtual learning environments depends not only on technology but also on usability, instructional planning, and proper integration within the educational system. These findings support the development of EmpowerEd by highlighting the importance of a well-structured and flexible LMS architecture that can accommodate students with disabilities.

Similarly, Alfonso, G. J. and Garcia, P. G. (2021) discussed Open and Distance e-Learning (ODEL) as an educational framework that combines pedagogy, educational technology, and e-learning systems to support flexible learning. Their work emphasized that LMS platforms are not only technological systems but also learning environments

that should encourage collaboration, learner engagement, accessibility, and student-centered instruction. The authors further noted that effective online learning depends on instructional design and flexible teaching strategies. This supports the goals of EmpowerEd because the system aims to provide an inclusive and accessible learning platform specifically designed for students with disabilities.

The acceptability and effectiveness of LMS platforms were further examined by Rogers, J. K. B., Salazar, R. P., and Buladaco, M. V. M. (2025) through a comparative study of Moodle and Google Classroom in Philippine higher education institutions. Using the Technology Acceptance Model (TAM), the study revealed that both platforms were considered effective and acceptable by students in terms of usefulness and ease of use. The researchers also highlighted that accessibility, institutional support, and user familiarity significantly influence the successful adoption of LMS platforms. These findings are relevant to EmpowerEd because they show that accessible and user-friendly systems improve the learning experience and encourage effective participation in digital education.

In addition, Garcia, Manuel B. (2025) evaluated the usability of the Canvas LMS across Progressive Web Apps (PWA) and native mobile platforms. The study found that students and teachers experience LMS platforms differently based on their specific roles and workflows. Students focused more on ease of access to learning materials and assignment submissions, while teachers prioritized monitoring and management functions. The research emphasized that usability, ease of learning, and user satisfaction are critical factors in determining the effectiveness of an LMS. The study also concluded that systems should accommodate different user needs to prevent technical barriers from

affecting learning outcomes. This is significant for EmpowerEd because students with disabilities require accessible and adaptive interfaces that support their specific learning and navigation needs.

The importance of accessibility and inclusive learning in digital education was also emphasized by Pegalajar-Palomino, M. C. (2022), who conducted a systematic review on mobile learning and inclusive education. The study explained that digital learning platforms improve accessibility, flexibility, and personalized learning opportunities for diverse learners, including students with disabilities. However, the research also identified challenges such as digital inequality and lack of accessibility-focused design. These findings reinforce the need for EmpowerEd to integrate accessibility features that ensure equal learning opportunities for students regardless of their physical or cognitive conditions.

Furthermore, Gupta, S. and Chen, J. (2025) discussed the role of Personalized Learning Environments (PLEs) in improving educational experiences. According to the authors, effective personalized systems use adaptive technologies, flexible user interfaces, and data-driven approaches to address individual learner profiles and learning needs. Their work emphasized that adaptive and student-centered systems improve learner engagement and knowledge retention, especially for learners with diverse academic and physical needs. This concept is relevant to EmpowerEd because the proposed LMS seeks to provide personalized accessibility features that can adjust according to the specific disabilities and learning requirements of students.

A more disability-focused approach was presented by Chavan, R. R., Patil, S. D., Patil, I. R., Patil, S. K., and Yadav, P. M. (2025) through the development of EduLife, a learning platform specifically designed for specially-abled students. The researchers integrated accessibility features such as screen readers, sign language modules, voice navigation, adjustable contrast, and simplified layouts to support students with visual, hearing, cognitive, and mobility impairments. The study demonstrated that accessible UI/UX design significantly improves system navigation and usability for students with disabilities. This directly supports the development of EmpowerEd because it validates the effectiveness of integrating assistive technologies and simplified interfaces into an LMS platform.

Lastly, Abonitalla, J. R. and Dejarme, J. J. (2025) investigated the relationship between technology-based learning tools and academic performance among junior and senior high school students in the Philippines. Their findings revealed that accessibility was the only factor that showed a significant positive relationship with learning outcomes compared to engagement and general usefulness. The study concluded that educational technologies are only effective when they provide equitable access and inclusive design for all learners. This strengthens the foundation of EmpowerEd by providing evidence that accessibility is a critical component in improving student performance and ensuring meaningful participation in online learning environments.

Inclusive Educational Technologies

The successful implementation of inclusive educational technologies is heavily reliant on foundational frameworks like Universal Design for Learning (UDL), which proactively aims to remove educational barriers and accommodate diverse learner needs from the outset. According to Priyadharsini and Sahaya Mary (2024), UDL serves as a vital catalyst for accelerating learning by addressing diverse cognitive styles, fostering collaboration, and creating a fundamental culture of equity in schools. To assist educators in operationalizing this concept, Takacs et al. (2022) emphasize the necessity of embedding multiple means of representation, action, and expression directly into digital curriculum design. Furthermore, as learning increasingly shifts to digital spaces, systematic reviews of online education emphasize that UDL-driven, evidence-based practices are absolutely critical for supporting students with disabilities, ensuring that remote learning platforms do not inadvertently create new accessibility barriers (Yang, Duha, & Luo, 2024).

As these digital frameworks evolve, Artificial Intelligence (AI) has emerged as a powerful, yet complex, tool for advancing inclusion. The U.S. Department of Education (2023) acknowledges AI's transformative potential to augment teaching capabilities and provide highly personalized learning interventions for students with special needs, though it also warns against the inherent risks of algorithmic bias and data unfairness. Echoing this need for cautious integration, Arantes (2025) explores AI governance through the lens of the social model of disability, arguing that the commercialization of AI-enabled assistive tools often exacerbates digital poverty, leaving students from lower socio-

economic backgrounds unable to access specialized support. Consequently, there is a pressing need for inclusive, participatory governance that centers the lived experiences of students with disabilities to ensure equitable access. Beyond broad frameworks and AI systems, highly specific assistive technologies play a direct, daily role in creating accessible classrooms. Asthana and Gupta (2025) highlight the profound impact of Automatic Speech Recognition (ASR), demonstrating that it empowers students with physical and learning disabilities by providing real-time transcription and facilitating hands-free note-taking. Similarly, for deaf and hard-of-hearing students, visual accommodations are vital. Pavani and Leonetti (2025) found that captions remain a highly effective tool for factual content access and procedural learning, even in educational environments where sign language interpretation is already available.

For all of these inclusive technologies to be effective, educators must be adequately trained and equipped to implement them. The Teacher Task Force (2020) explicitly notes that assistive technologies ranging from high-contrast settings to built-in speech-to-text software should be viewed as universal resources that enhance independence for all students, rather than isolated tools for a few. Finally, the long-term effectiveness of these classroom interventions relies on continuous evaluation, with the Assistive Technology Industry Association (2024) emphasizing the importance of rigorously tracking the real-world outcomes and benefits of assistive technologies to continuously advance evidence-based practices and improve overall service delivery for students.



Global Perspective and Implementation

The laws and policies that are disability inclusive on the global scale provide a critical base for substantive equality in education. The UN Convention on the Rights of Persons with Disabilities (CRPD) breaks away from the medical deficit model and calls for a rights-based approach to education, including an inclusive education system at all levels for states. A detailed examination of all national laws in 193 countries, however, shows that the national commitments are not yet fully harmonised; 46% of the countries have not yet extensively prohibited disability-based discrimination in secondary education; 35% do not lawfully guarantee access to mainstream environments with necessary individual accommodations (Waisath et al., 2024).

In the Philippine context, RA 10754 is an important mechanism to promote equity at the national level. This law extends the benefits and privileges of Persons with Disabilities (PWDs) to help eliminate the economic and social barriers which frequently stand in the way of students to access the educational tools and institutions. This is a legislative step towards inclusion, but there is a social and ethical aspect to it. Unfortunately, the stigma is still present, and there is no granular enforcement mechanism existing yet, which indicates that there is still a need for a wider cultural shift to achieve true inclusiveness if legislative changes are to be made (Ahmed, 2024). These national policies need to be implemented at the institutional level in the form of a roadmap, such as the Alternative Learning System (ALS) Strategic Plan of the Department of Education for the inadequate learners who are not attending school. The Department of Education (DepEd, 2022) provides a systematic blueprint to increase access to basic education,

including those of out-of-school youth and adults with special needs, without being left behind. To be effective on the ground, however, the national strategies need to be translated into local knowledge sharing processes, which must be facilitated by academic institutions. Inclusive knowledge can be delivered by institutional repositories and indigenous localized resources, as is explored in the regional academic initiative, the E-Dawa project at Ilocos Sur Polytechnic State College (ISPSC), 2021, where the gap in implementing inclusive knowledge from the national level to the community level is evident.

Teachers play a pivotal role in implementing these inclusive legal and institutional frameworks for success. While educators help to realize student potential by identifying and leveraging students' strengths, the effectiveness of their work is influenced by how well trained they are and whether they have access to structured support (Baidya & Baidya, 2025). Teacher Support Programmes (TSP) have been developed as valuable guides between theory and practice to provide six-stage models to assist regular teachers in dealing with the specific concerns of regular teachers supporting students with disabilities (Mehmood & Parveen, 2025).

In addition to pedagogy, the use of new technology is becoming the basis of inclusive life. Today, intelligent systems, such as Artificial Intelligence (AI), the Internet of Things (IoT) and advanced robotics, are being used to address physical and cognitive challenges. Such technologies can provide tailored support, like smart navigation for the visually impaired or AI communication systems, improving the independence and lifestyle of people with disabilities (Setiawan, 2024). AI has the potential to create highly adaptive

learning environments, deliver real-time feedback, and offer customized scaffolding for diverse learners, as highlighted by the Office of Educational Technology (OET), which also warns that the use of AI in education needs to be guided by ethical principles to prevent algorithmic bias and safeguard student data privacy (U.S. Department of Education, 2023).

In low and middle-income countries (LMICs), introducing such educational technology (EdTech) comes with its own set of challenges. The findings from systematic reviews suggest that EdTech can have a significant impact on enhancing learning outcomes for primary school learners with disabilities, but the effectiveness of these interventions is strongly linked to the access to inclusive software, teacher digital literacy training, and addressing infrastructures challenges such as internet access (Lynch et al., 2024). Global digital governance frameworks focus on standardization as a way to overcome these barriers. Handbooks on web and digital accessibility are useful toolkits for developers and educators to help build universally designed and accessible digital learning spaces, with clear, guidelines for creating text alternatives, keyboard accessibility, and screen-reader compatibility (Digital Policy Office, 2024).

Lastly, setting up curricula and assistive technology is essential for lifelong independence. Mastam and Zaharudin (2024) developed a framework unique in focusing on work-readiness skills including communication and self-advocacy skills for vocational students. As a result of the use of appropriate accommodation and assistive technology, students will be better able to carry out job duties efficiently. To overcome the employment



barriers and encourage diversity in the world of work, this inclusive approach will involve teacher support, intelligent systems, and vocational training (Mastam & Zaharudin, 2024).

aims to surmount employment obstacles and promote diversity within the global labor market (Mastam & Zaharudin, 2024).

Synthesis of Review of Related Literatures

Reviewing the related literature and studies highlights a critical technological gap in modern learning ecosystems. While general Learning Management Systems (LMS) offer robust tools for content distribution and basic classroom organization, they are fundamentally built on static interface frameworks. This architectural rigidity forces students with special educational needs to rely heavily on fragmented third-party hardware extensions or external screen-reading software. These external workarounds frequently break during web navigation, fail to parse rich multimedia elements accurately, and significantly increase the cognitive load of the learner.

Furthermore, existing academic tracking systems operate in isolation from the client-side user experience. Current platforms log performance metrics, such as quiz scores and submission timestamps, but fail to map how specific digital barriers influence those performance outcomes. This data gap prevents educators from identifying whether a poor submission is caused by a student's misunderstanding of the academic content or a failure to navigate an inaccessible digital environment.

EmpowerEd directly addresses these systemic limitations and serves as an integrated solution rather than a duplication of existing software. The project unifies client-side accessibility tools with a relational backend tracking architecture. Instead of forcing



a student to adapt to a fixed layout, the platform evaluates user-defined disability profiles (visual, hearing, or learning needs) and dynamically shifts its styling layers and media renderers. It embeds custom high-contrast styling engines, native text-to-speech modules, and synchronized media captions directly into the core system layout. This provides educators with a consolidated monitoring dashboard that correlates a student's performance directly with the accessibility tools they utilize. By integrating custom, profile-driven user interfaces with an empirical research data repository, EmpowerEd closes the gap between digital accessibility and educational data diagnostics, establishing a verifiable baseline for inclusive e-learning system architectures.

CHAPTER 3

RESEARCH METHODOLOGY

Introduction

The methodology chapter outlines the systematic process utilized to design, develop, and evaluate EmpowerEd: Think. Learn. Achieve— an accessible online learning platform for students with disabilities. By integrating the Universal Design for Learning (UDL) framework, this study seeks to proactively eliminate educational barriers through specialized web features like text-to-speech, adjustable font sizes, and video subtitling. This chapter presents the research methodology used in the study, including the quantitative research design, purposive sampling technique for participant selection, and the technical architecture employed in the development of the platform to evaluate its effectiveness in improving academic performance.

Research Methodology

The proponents utilize a Quasi-Experimental Research Methodology, specifically the Pre-test/Post-test Control Group Design, to evaluate the effectiveness of the EmpowerEd platform. This methodology is defined as a systematic process used to identify and analyze the impact of a specific intervention, in this case, an accessible online learning platform, on a target group without the use of random assignment. For this study, the methodology encompasses the following key aspects:



- A. **Research Design:** The study employs a quasi-experimental design involving two distinct groups: an Experimental Group (N = 30) and a Control Group (N = 30).

This design is chosen because it allows for a direct comparison between students using the EmpowerEd platform and those using traditional learning methods to determine if the software significantly improves academic performance. To guide the technical execution of this design, the proponents will combine the Agile Software Development Methodology, Scrum framework, and the Prototype Model to allow continuous improvement and client feedback during system development. Operating within this agile setup, the software development lifecycle is broken down into manageable, two-week sprint cycles driven by continuous user-interface evaluations. Furthermore, the researchers integrate the Prototype Model into these sprints to build functional, early versions of the platform's core accessibility features—such as text-to-speech engines, adjustable font layouts, and video captioning tools. This combined approach ensures that the unique human-computer interaction (HCI) and sensory needs of students with visual, hearing, and learning disabilities are tested, analyzed, and refined by actual users before the final web system is fully deployed.

- B. **Data Collection Methods:** The primary procedure follows a four-stage process:

1. A pre-test to establish a performance baseline for both groups.
2. An Intervention Phase where the experimental group utilizes EmpowerEd's accessibility features (text-to-speech, captions, etc.) for a specific period.
3. A post-test administered to both groups to measure learning gains.



4. A System Evaluation Survey given to the experimental group to gather quantitative feedback on usability.
- C. **Research Instrument:** Two main instruments are utilized:
- a) Standardized Achievement Tests (Pre-test and Post-test) to gather performance data.
 - b) A Structured Survey Questionnaire using a 5-point Likert Scale to measure user satisfaction and the effectiveness of specific accessibility tools.
- D. **Data Analysis Methods:** The researchers will apply statistical analysis, such as T-tests, to compare the mean scores of the pre-test and post-test results. This allows the team to mathematically determine if the platform had a significant effect on learning, thereby testing the study's null (H_0) and alternative (H_1) hypotheses.

Formula

$$t = \frac{\bar{d}}{s_d/\sqrt{n}}$$

Where:

$\bar{d}$ = mean of the differences (post-test – pre-test)



s_d = standard deviation of the differences

n = number of paired observations

Degrees of Freedom

$$df = n - 1$$

- E. **Overall Framework:** The research is conducted within an inclusive education framework that seeks to dismantle digital barriers. By combining system performance data with user feedback, the proponents ensure that the platform effectively bridges the digital divide and promotes equal access to education for students with disabilities.



Software/Hardware Requirements

This section presents an overview of the technical architecture required to build and run the EmpowerEd platform. It details the specific software systems, programming frameworks, database structures, and physical hardware components necessary for development and deployment. Each component is listed along with a brief discussion of its operational role and purpose in ensuring a stable, accessible learning environment.

Table 1: Software Requirements for Systems Development

Software Classification	Software Name
Integrated Development Environment (IDE)	Visual Studio Code
Programming Languages and Frameworks	HTML5
	CSS3
	JavaScript
	Node.js
	Express.js
Database	MongoDB
Third-Party Libraries	Font Awesome
	Google Fonts



	Bootstrap 5
	CSS Flex Box
Others	Git, GitHub, Windows 11, Google Chrome, WCAG Checker

Visual Studio Code: Functions as the primary Integrated Development Environment (IDE) used by developers to write, format, debug, and manage frontend layouts and backend server-side logic.

HTML5, CSS3, and JavaScript: Form the core client-side technology stack, managing semantic document layouts, responsive web styles, and interactive interface behaviors.

Node.js & Express.js: Serve as the backend execution environment and web framework, managing client-server requests, secure API routing modules, and database connection pools.

MongoDB: Functions as the primary document-oriented NoSQL database system, storing dynamic student accessibility profiles, learning material links, and operational data.

Third-Party Web Utilities: Includes design frameworks like Bootstrap and Flexbox alongside asset systems like Font Awesome and Google Fonts to generate highly scannable, accessible, and structured user interfaces.



Git & GitHub: Utilized as the mandatory version control systems to track code evolution, document changes, and host the secure repository system.

Windows 11: Serves as the primary operating system environment for the development team, providing native compatibility for running host servers, command-line interfaces, and execution environments.

Google Chrome: Functions as the primary client-side execution platform used to render, test, and inspect the application's HTML on both mobile web browser and PC web browser.

WCAG Checker: Website used to verify the system's colors and features are in compliance with the Web Content Accessibility Guidelines (WCAG).

Table 2:Hardware Requirements for Systems Development

Hardware Component	Technical Specifications
Central Processing Unit (CPU)	Intel Core i5 / AMD Ryzen 5 (6-Core, 3.0 GHz or higher)
Random Access Memory (RAM)	16 GB DDR4 / DDR5
Storage Architecture	512 GB Solid State Drive (SSD)
Mobile Testing Device	Android Smartphone (Minimum 6.0" Display, 4GB RAM)



Peripherals & Assistive Tools	Audio Playback Hardware (Headphones/Speakers)
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Central Processing Unit (CPU): Provides the multi-threaded computational processing power required to run active development servers, compile web frameworks, and run local databases concurrently.

Random Access Memory (RAM): Ensures the system handles intensive background processes, virtual terminals, and complex editing environments without experiencing fatal slowdowns.

Storage Architecture (SSD): High-speed solid-state drive baseline to optimize compilation speeds, deployment workflows, and quick access to large local assets.

Mobile Testing Device: A physical smartphone environment used to test responsive scaling rules, real-world button responsiveness, and visual accessibility rendering performance.

Peripherals & Assistive Tools: Mandatory audio components used to audit accessibility compliance, including text-to-speech speed intervals and closed-caption timing synchronization.



Functional Requirements

The functional requirements define the specific behaviors, features, and operations that the EmpowerEd web system must perform to serve both the students and educators effectively.

Table 3:Functional Requirements of the Project

FR No.	Module No.	Function Name(s)	Description
FR-01	WEB-01	Log in Authentication System	The system allows users to securely log in their account and transfers them to their respective dashboards (Student or Teacher)
FR-02	WEB-02	Accessibility Panel	The system provides a easy to use accessibility panel allowing users to adjust font sizes (Small, Medium, or Large) and toggle High Contrast mode across all pages
FR-03	WEB-02	Course Dashboard	The system displays a dashboard for students to access different modules about the current subject.
FR-04	WEB-03	Announcements	The system displays a dashboard to view real-time announcements posted by educators.



FR No.	Module No.	Function Name(s)	Description
FR-05	WEB-04	Course Module	The system displays accessible course materials and lessons about the current subject.
FR-06	WEB-05	Video Module	The system provides an embedded media player with adjustable speeds, synchronized subtitles, and scrollable text transcripts.
FR-07	WEB-06	Assessment Module	The system allows students to take interactive quizzes and activities, automatically calculating and recording scores using a standard academic grading system where 4.0 is the highest achievable mark.
FR-08	WEB-07	Assignment Submission	The system provides an interface for students to upload and submit required output for specific tasks.
FR-09	WEB-02	Progress Tracking	The system tracks user completion across all modules, dynamically updating a visual progress bar.
FR-10	WEB-07	Educator Dashboard	The system enables teachers to upload multi-modal materials, create assessments, post



FR No.	Module No.	Function Name(s)	Description
			announcements, and monitor quantitative student data.



Non-Functional Requirements

The non-functional requirements describe the EmpowerEd system's operation capabilities, constraints, and quality attributes such as usability, performance, portability, data integrity, and security. Table 7 details these specifications.

Table 4: Non-Functional Requirements of the Project

Non-Functional Criteria	Description
Usability	The system should: 1. Adhere strictly to Web Content Accessibility Guidelines (WCAG) to ensure the interface is inclusive. 2. Maintain a flat, consistent navigation structure across all modules to benefit users with cognitive or learning disabilities. The system is constrained to: 1. Limit the use of flashing or highly dynamic UI elements to prevent overwhelming users with cognitive or learning disabilities.



Non-Functional Criteria	Description
Performance	The system should: 1. Fetch and render dynamic resources (like .txt transcripts and quiz data) asynchronously to prevent UI freezing. The system is constrained to: 1. Maintain stable functionality and acceptable loading times even on low-bandwidth internet connections..
Portability	The system should: 1. Utilize responsive design frameworks (e.g., Bootstrap) to remain fully accessible and functional across desktop, tablet, and mobile web browsers. The system is constrained to: 1. Operate entirely within standard web browsers without requiring the installation of specialized third-party plugins or external applications.
Data Integrity	The system should:



Non-Functional Criteria	Description
	1. Implement automated logging of all task submissions and video completion milestones to prevent accidental data loss. The system is constrained to: 1. Accurately store performance grading and completion statuses to ensure the statistical validity of data.
Security	The system should: 1. Encrypt user authentication credentials and safely manage active login sessions. The system is constrained to: 1. Comply with strict data privacy protocols to protect the identities, disability categories, and performance metrics of the SPED students. 2. Restrict access to specific modules (e.g., grade viewing or content uploading) based on the authenticated user's role (Student vs. Teacher).



Sample Description

The study will involve a total of 60 Grade 4 students with “Mid to High” and “High” IQ levels currently enrolled in Special Education (SPED) programs or inclusive education settings for the academic year 2025-2026. These participants are selected based on their specific needs for digital accessibility tools to evaluate the effectiveness of the EmpowerEd platform.

Sampling Technique

The researchers will utilize Purposive Sampling. This non-probability sampling method is employed to ensure that the respondents possess the specific characteristics required to test the platform’s features. Participants are specifically chosen because they require audio support, visual aids, or simplified navigation to access educational content.

Participant Classification

The sample is categorized based on the disability profiles that the EmpowerEd platform aims to support:

- **Visual Disabilities:** Students who require audio support (Text-to-Speech), benefits from high-contrast modes, or adjustable text sizes to perceive learning materials.
- **Hearing Disabilities:** Students who require visual aids and synchronized captions for video-based lessons.
- **Learning Disabilities:** Students who benefit from simplified system navigation and repeated access to downloadable materials to maintain learning focus.

Research Groups

For the purpose of quantitative comparison and statistical analysis, the 60 participants are divided into two distinct groups:

1. **Experimental Group (30 Students):** This group will utilize the EmpowerEd platform for their lessons and school activities to test its specialized accessibility features.
2. **Control Group (30 Students):** This group will serve as the baseline for comparison. They will not use the platform and will instead continue with their current learning methods, such as printed modules or standard websites.

Data Collection Procedures

The data collection process for the study titled “EmpowerEd: Think. Learn. Achieve. – Developing an Accessible Online Learning Platform for Students with Disabilities” will follow a systematic procedure to ensure the accuracy and reliability of the gathered data.

First, a formal letter of request will be sent to the school principal, Special Education (SPED) coordinator, and teachers handling students with disabilities to seek permission to conduct the study and gather data from the selected participants. After approval is granted, the researchers will coordinate with the assigned teachers to identify qualified participants who meet the criteria of the study.



The study will involve 60 students with disabilities from SPED programs or inclusive learning environments. These participants will be divided into two groups: 30 students for the experimental group and 30 students for the control group. The experimental group will use the EmpowerEd platform, while the control group will continue using their usual learning methods such as printed modules, teacher-provided materials, or standard online platforms.

Before the implementation of the platform, both groups will be given a pre-test to determine their initial level of learning performance. The pre-test will cover the selected lesson topics and will serve as the basis for comparison after the intervention.

After the pre-test, the experimental group will be introduced to the EmpowerEd platform and oriented on how to use its accessibility features, including text-to-speech, subtitles, adjustable font sizes, and downloadable learning materials. The group will use the platform for a specific period, such as two to four weeks, during their regular lessons and assigned activities. During this period, teachers will upload lessons, provide tasks, and monitor the students' progress through the system. Meanwhile, the control group will continue their regular learning process without using the EmpowerEd platform for the same duration.

At the end of the implementation period, both groups will be given a post-test similar in content and level of difficulty to the pre-test. The results of the post-test will be collected and compared with the pre-test scores to determine whether the use of the EmpowerEd platform significantly improved the learning performance of students with disabilities.

In addition, feedback from both the students and teachers in the experimental group may be gathered through a survey questionnaire or interview to assess the usability, accessibility, and effectiveness of the platform. The collected data will then be tabulated, analyzed, and interpreted using appropriate statistical tools.

Data Privacy and Security

The researchers will strictly comply with the provisions of Data Privacy Act of 2012 to ensure the protection and proper handling of participants' personal information. Before gathering data, informed consent will be obtained from the participants, their parents or guardians, and the school administration involved in the study.

All collected information, including test results, survey responses, interview data, and system records, will be treated with strict confidentiality and will only be used for academic and research purposes. Personal information such as names and identities will not be disclosed in any part of the study. Instead, codes or pseudonyms will be used during data analysis and presentation of results.

To protect the gathered data, all digital files and records will be stored in password-protected devices and secure storage systems accessible only to the researchers. The collected data will not be shared with unauthorized individuals or third-party organizations.

Ethical Considerations

The researchers will observe ethical standards throughout the conduct of the study. Participants will be informed about the purpose of the research, the procedures involved, and their right to withdraw from the study at any time without penalty.



The study will ensure respect, fairness, and inclusivity for all participants, especially students with disabilities. The researchers will avoid any form of discrimination, bias, or unfair treatment during the implementation of the study. Participation in the study will remain voluntary, and all gathered information will be handled responsibly and ethically.

Responsible AI Use

The researchers acknowledge the importance of responsible and ethical practices in the development and implementation of web-based educational systems. The EmpowerEd platform primarily functions as an accessible online learning system and does not rely on artificial intelligence (AI) or machine learning technologies for its core operations. However, the researchers utilized AI tools during the documentation, content development, grammar checking, and system planning processes to assist in improving clarity, organization, and productivity. All AI-generated outputs were carefully reviewed, validated, and revised by the researchers to ensure accuracy, fairness, and the absence of biased or misleading information.

The researchers will ensure that the platform promotes fairness, inclusivity, accessibility, and equal learning opportunities for all users. The system will be used solely for educational purposes and under the supervision of teachers and school administrators.

Furthermore, the researchers will ensure that all system processes and user data are handled responsibly and ethically in compliance with the Data Privacy Act of 2012. Sensitive information collected from users will be protected through proper data handling



procedures, confidentiality measures, and restricted access to authorized individuals only, ensuring the rights, privacy, and welfare of all users are safeguarded throughout the study.

System Design

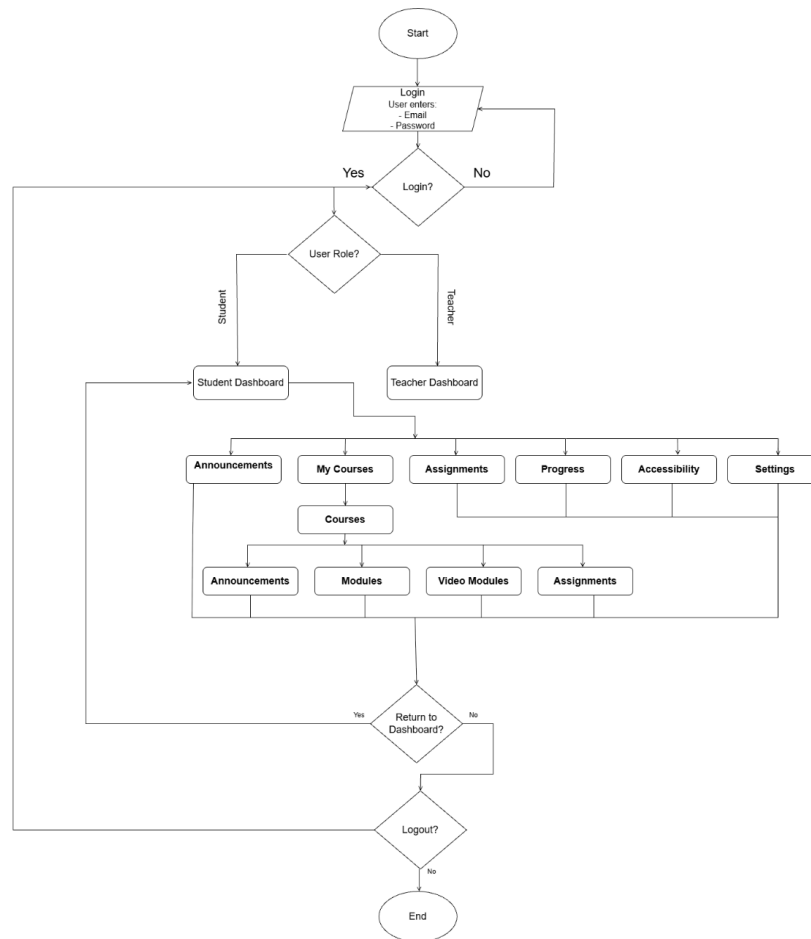


Figure 2:Flowchart

Flowchart

Figure 2 represents the flowchart of the EmpowerEd system illustrates the overall process and navigation of the accessible online learning platform for students and teachers. The process begins with user login, where the system verifies the entered email and password. Once authenticated, the user is directed to either the Student Dashboard or Teacher Dashboard based on their role. From the dashboard, users can access features such as announcements, courses, assignments, progress tracking, accessibility



settings, and account settings. Within the courses section, users can view modules, video modules, assignments, and announcements. The flowchart also shows the options to return to the dashboard or log out of the system, which ends the session.

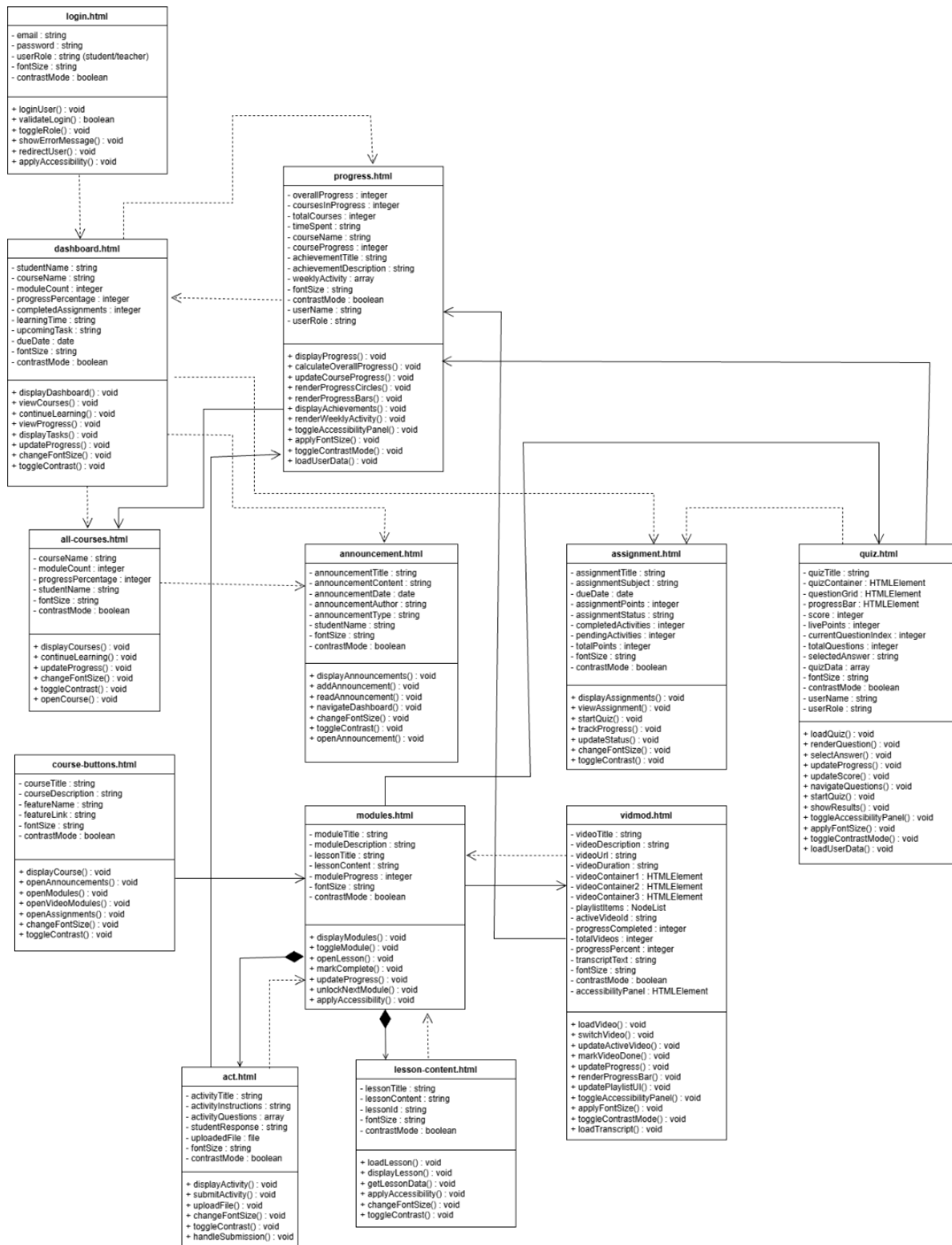


Figure 3: Unified Modeling Language



Unified Modeling Language

Figure 3 shows the UML of the EmpowerEd system. It presents the structure and relationship of the different web pages and components of the accessible learning management system. It shows how the system is organized into modules such as login, dashboard, courses, announcements, assignments, quizzes, progress tracking, video modules, lessons, and activities. Each class contains attributes and functions that define the features and operations of the system. The diagram also illustrates the interaction between pages, allowing users to navigate through learning materials, monitor progress, complete assessments, and access accessibility features such as font size adjustment and contrast mode. Overall, the diagram provides a clear representation of the system's functionality and connections between components.

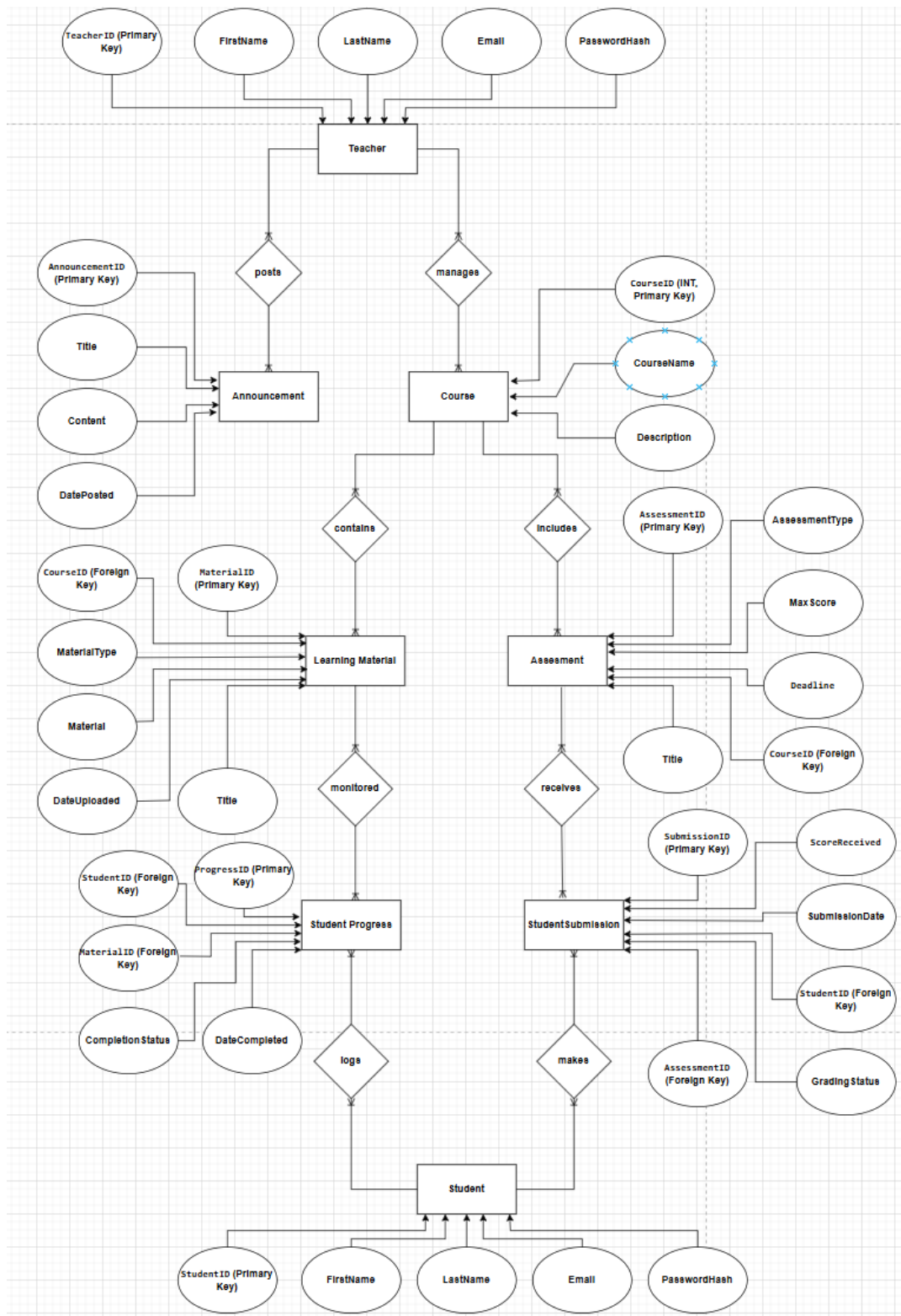


Figure 4: Entity Relationship Diagram

Database design/ERD

Figure 4 represents how the EmpowerEd database is structured. Interacting around 3 main tables TEACHER, STUDENT, and COURSE. The TEACHER table connects to ANNOUNCEMENT and COURSE to handle classroom content. The COURSE table connects to both the Learning Materials and Assessments while the STUDENT_PROGRESS tracks viewed learning materials and STUDENT_SUBMISSION to manage turned-in assignments.

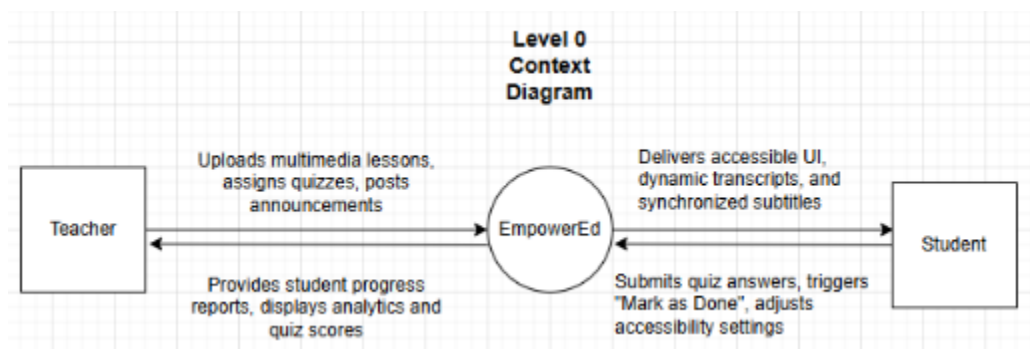


Figure 5: Data Flow Diagram – Level 0: Context Diagram

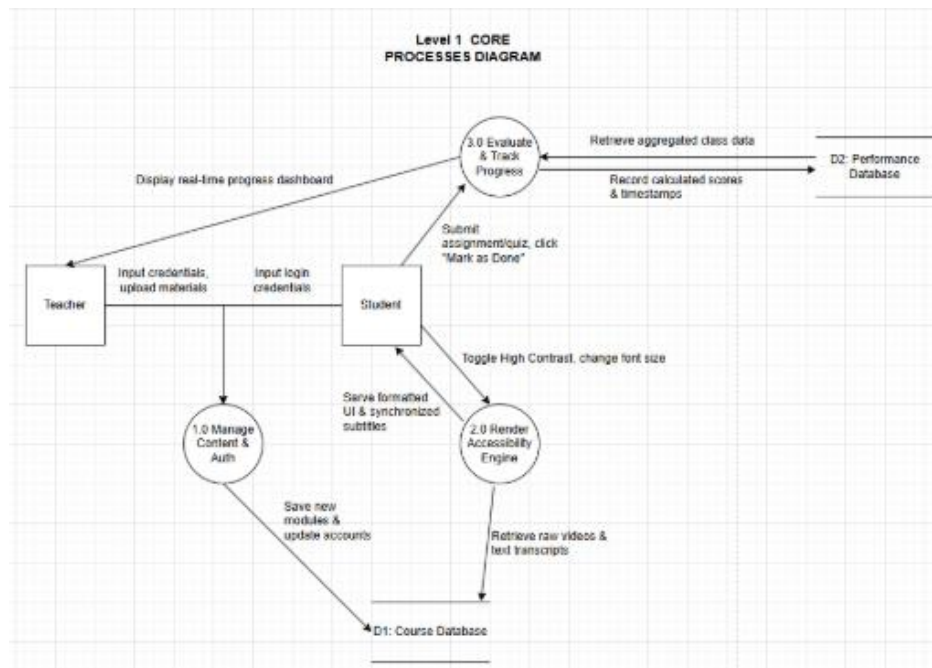


Figure 6: Data Flow Diagram – Level 1: Core Process Diagram

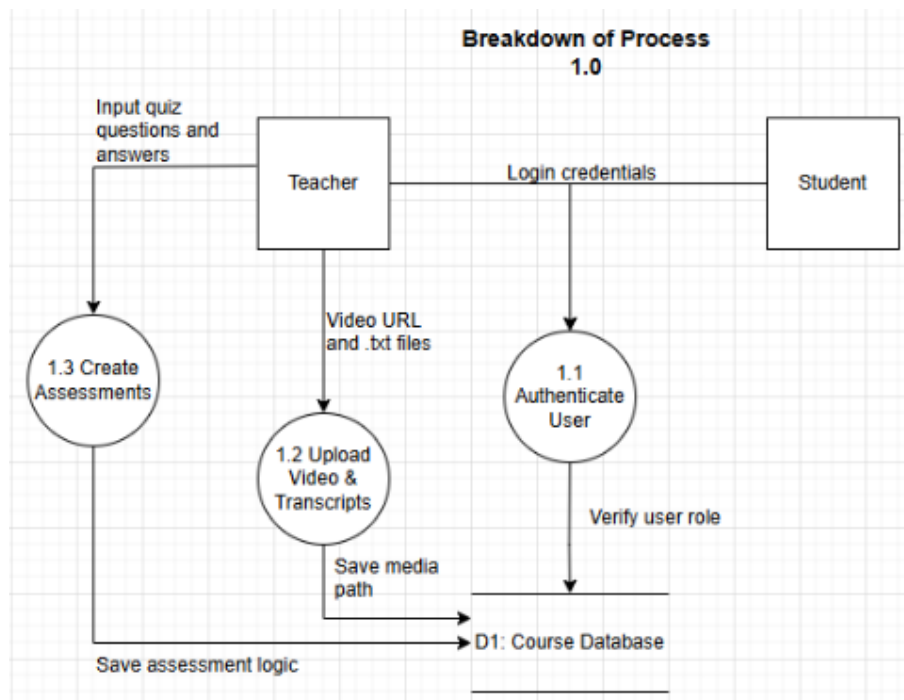


Figure 7: Data Flow Diagram – Level 2: Breakdown of Process 1.0

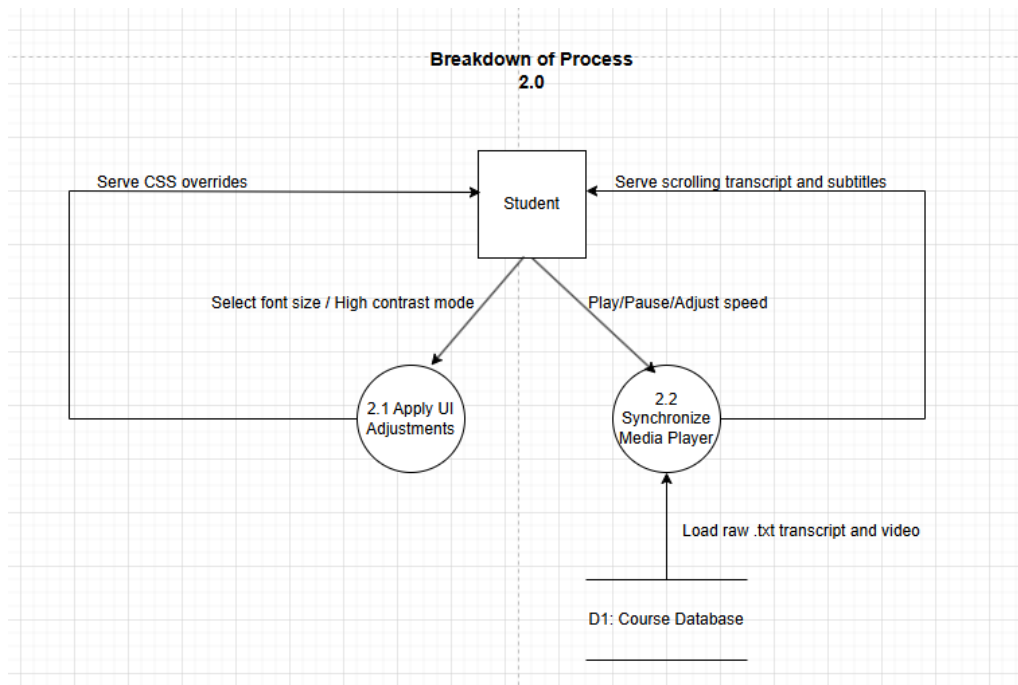


Figure 8: Data Flow Diagram – Level 2: Breakdown of Process 2.0

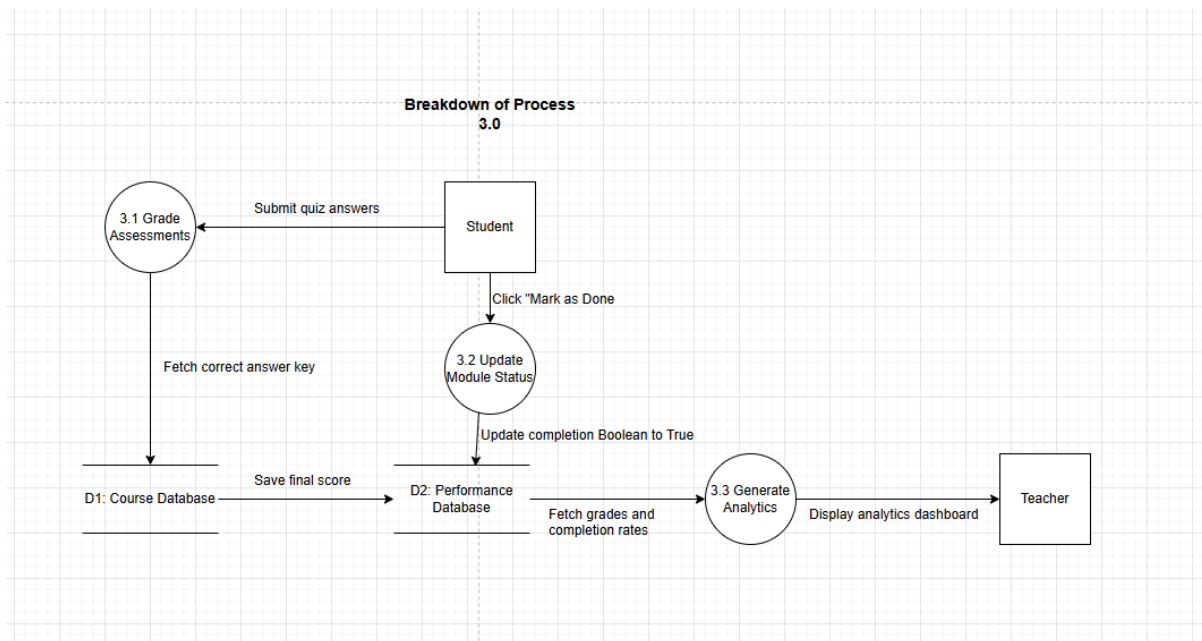


Figure 9: Data Flow Diagram – Level 2: Breakdown of Process 3.0

Data Flow Diagram

Figure 5-9 represents how the data then information moves through and the different modules of the EmpowerEd system. The Level 0 Context Diagram provides a high-level view, showing how teachers upload lessons and assignments while students log their progress, submit quizzes, and change accessibility options. Shifting to a deeper view, the Level 1 Diagram breaks this down into three main processes: 1.0 Manage Content & Auth, 2.0 Render Accessibility Engine, and 3.0 Evaluate & Track Progress.

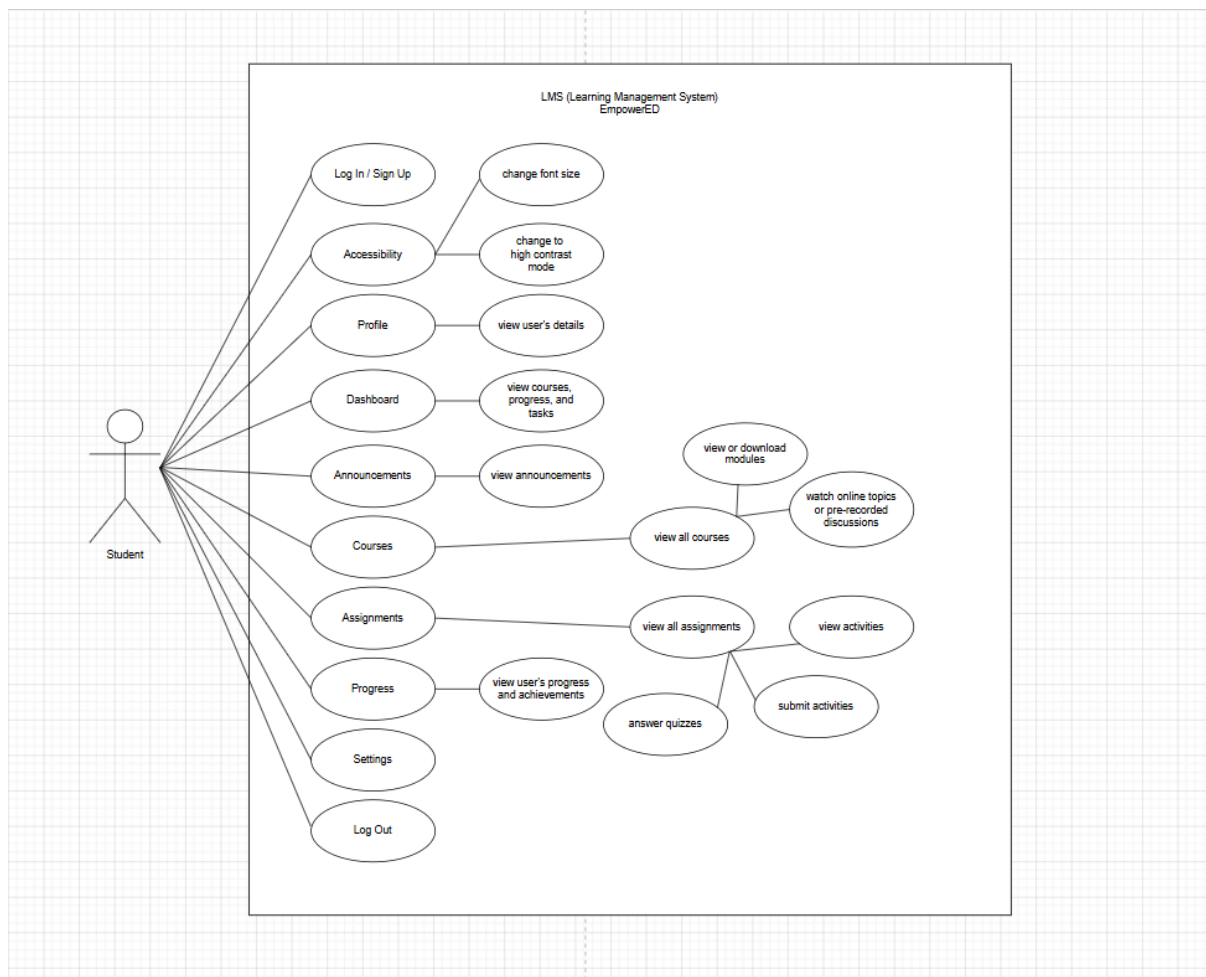


Figure 10: Use Case Diagram

Use Case Diagram

Figure 10 represents the Use Case Diagram of EmpowerEd. It illustrates the functional capabilities provided to the student actor within the EmpowerEd Learning Management System (LMS). The system architecture is categorized into three primary functional areas:

- **Core LMS Functions:** Provides standard portal operations, including account authentication (Log In / Sign Up / Log Out), announcement tracking, and profile management.
- **Accessibility Layer:** Integrates built-in accessibility configurations directly into the interface, specifically dynamic font scaling and a high-contrast mode toggle to accommodate users with physical, visual, or cognitive disabilities without relying on third-party plugins.
- **Academic and Progress Management:** Enables autonomous learning workflows by allowing students to access a centralized Dashboard to track tasks, view or download modules for offline flexibility, stream captioned lecture videos, answer digital quizzes, and submit academic activities.

SYSTEM ARCHITECHTURE

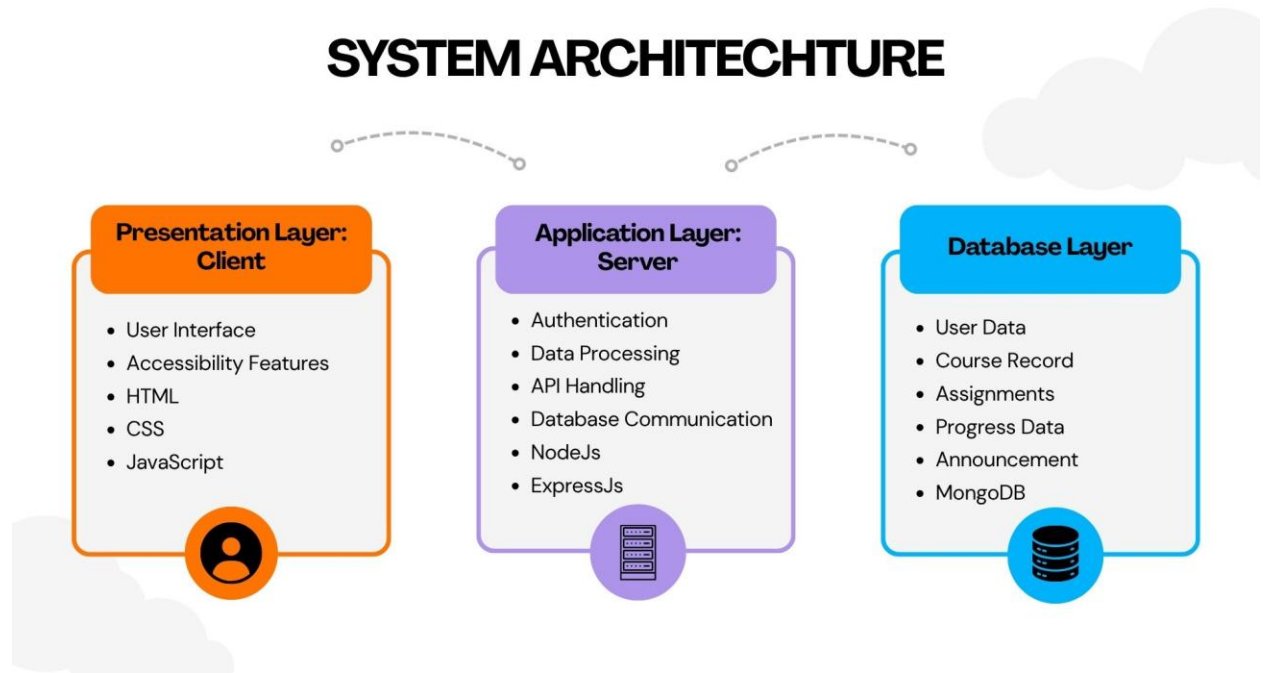


Figure 11: System Architecture

System Architecture

Figure 11 illustrates the system architecture of the proposed EmpowerEd Learning Management System, which is composed of three main layers: the Presentation Layer, Application Layer, and Database Layer. These layers work together to provide an accessible and efficient learning platform by handling user interaction, processing system operations, and securely storing essential data such as user information, assignments, course records, and progress reports.

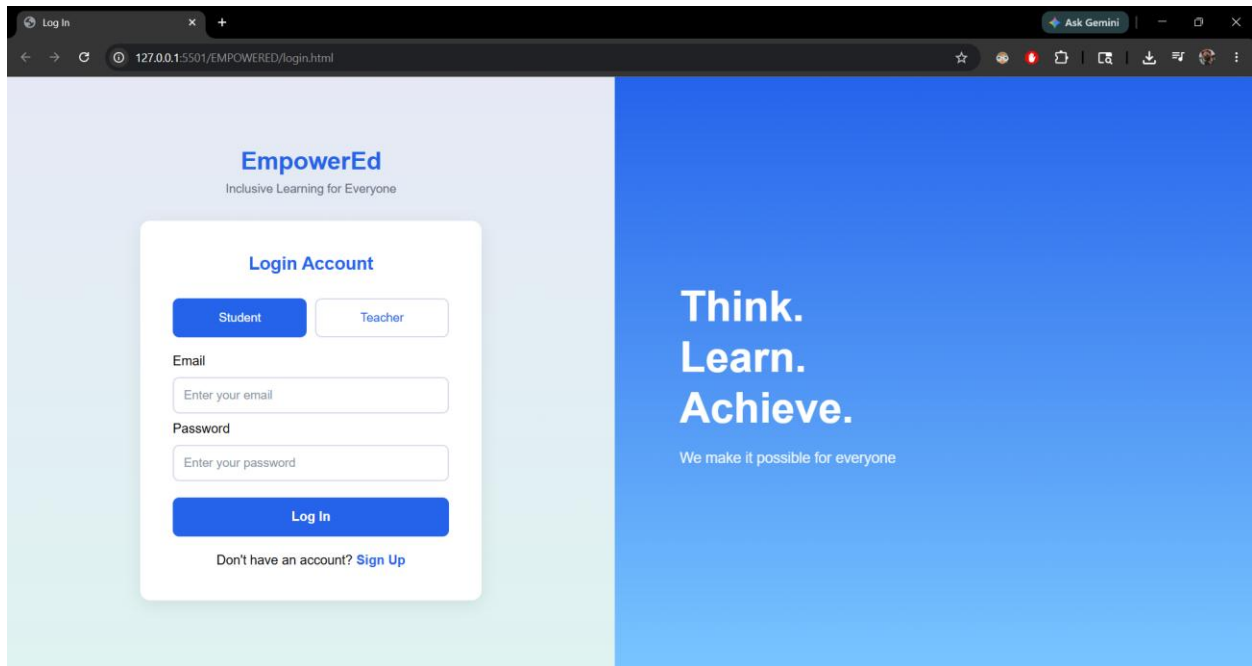


Figure 12: Log In Page

User Interface: Log in Page

Figure 12 presents the log in page which allows users—students or teachers to log in into their EmpowerEd account. The page is designed to be user-friendly and accessible to provide smooth and convenient login experience for all users, especially students with disabilities.

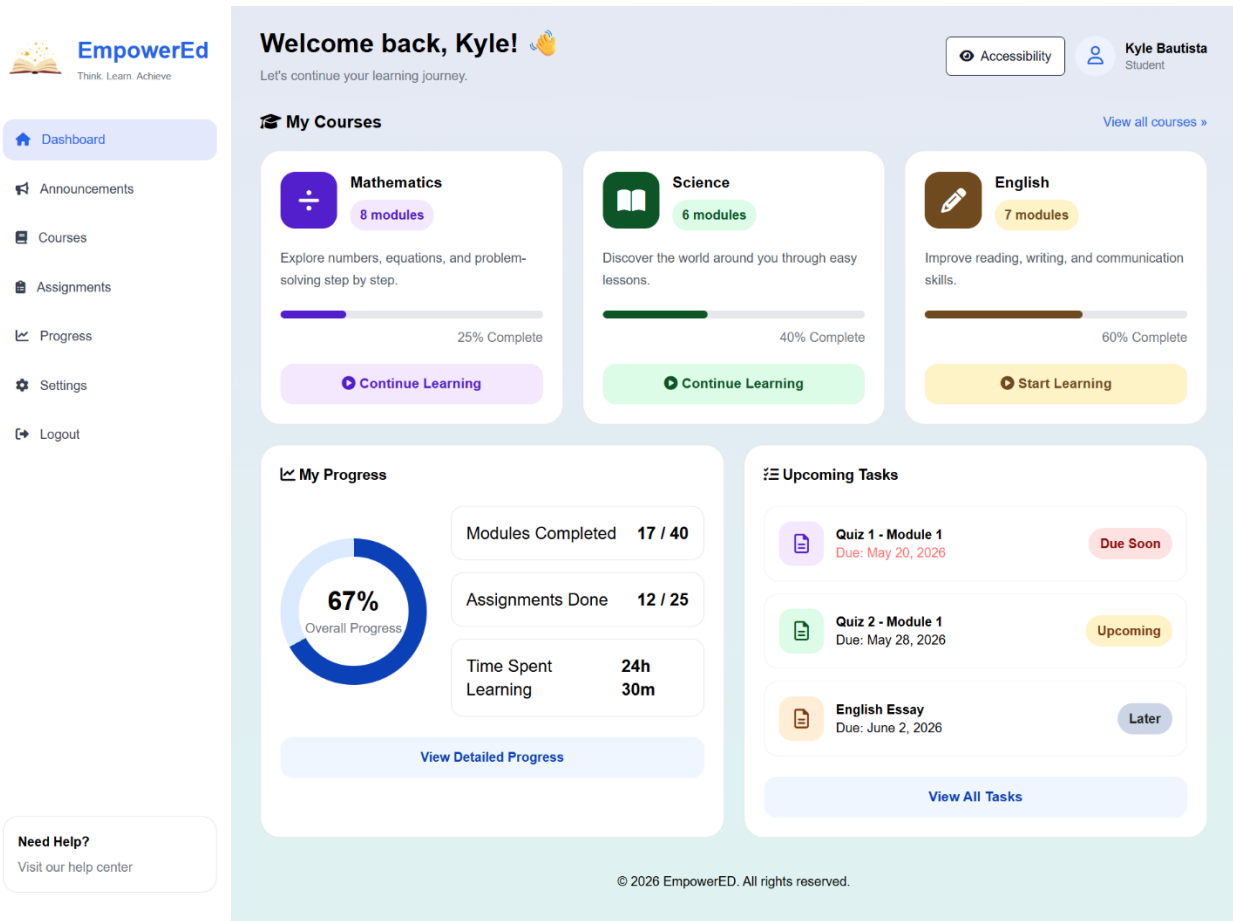


Figure 13: Dashboard Panel

User Interface: Dashboard Panel

Figure 13 shows the dashboard panel which serves as the main interface. This is where users can view their recent courses, summary of their progress, and upcoming tasks. They can also access several buttons on the sidebar which serves as navigation to other modules. The dashboard is designed to provide an organized, intuitive, and accessible learning environment for the students.


EmpowerEd
Think. Learn. Achieve.

Dashboard
Announcements
Courses
Assignments
Progress
Settings
Logout

Accessibility
Kyle Bautista
Student

Announcements

Stay updated with the latest announcements.

📌 **Course Extension: Assignment Deadline Updated**
Read Aloud

Dr. Joachim Barboza
May 20, 2026
Important

Due to technical difficulties, Assignment 3 has been extended to April 27, 2026.

📌 **New Accessibility Resources Available**
Read Aloud

Dr. Alessandra Metrillo
April 18, 2026

New screen reader tools and accessibility resources are now available in Module 2.

📌 **Quizzes Now Available: Module 2 & 3 Assessment**
Read Aloud

Dr. Ken Gonzales
April 24, 2026

Module 2 and Module 3 quizzes are now uploaded. Please check Pending Activities.

📌 **Final Project Guidelines Reminder**
Read Aloud

Dr. Neomi Gabejan
April 25, 2026
Important

Please be informed that the Final Project Guidelines have now been released. Kindly take time to carefully read and understand all the requirements, instructions, and deadlines provided to ensure proper compliance.

📌 **Peer Review Activity Deadline Reminder**
Read Aloud

Dr. Kyle Bautista
April 26, 2026

All students must complete the peer review activity before the deadline. Late submissions will not be accepted.

Need Help?
Visit our help center

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Figure 14: Announcement Module

User Interface: Announcement Module

Figure 14 shows the announcement module which allows the students to view announcement from their teachers. The developers included “read aloud” button that enables student to listen to the announcements. This is designed for students with visual impairments and reading difficulties.

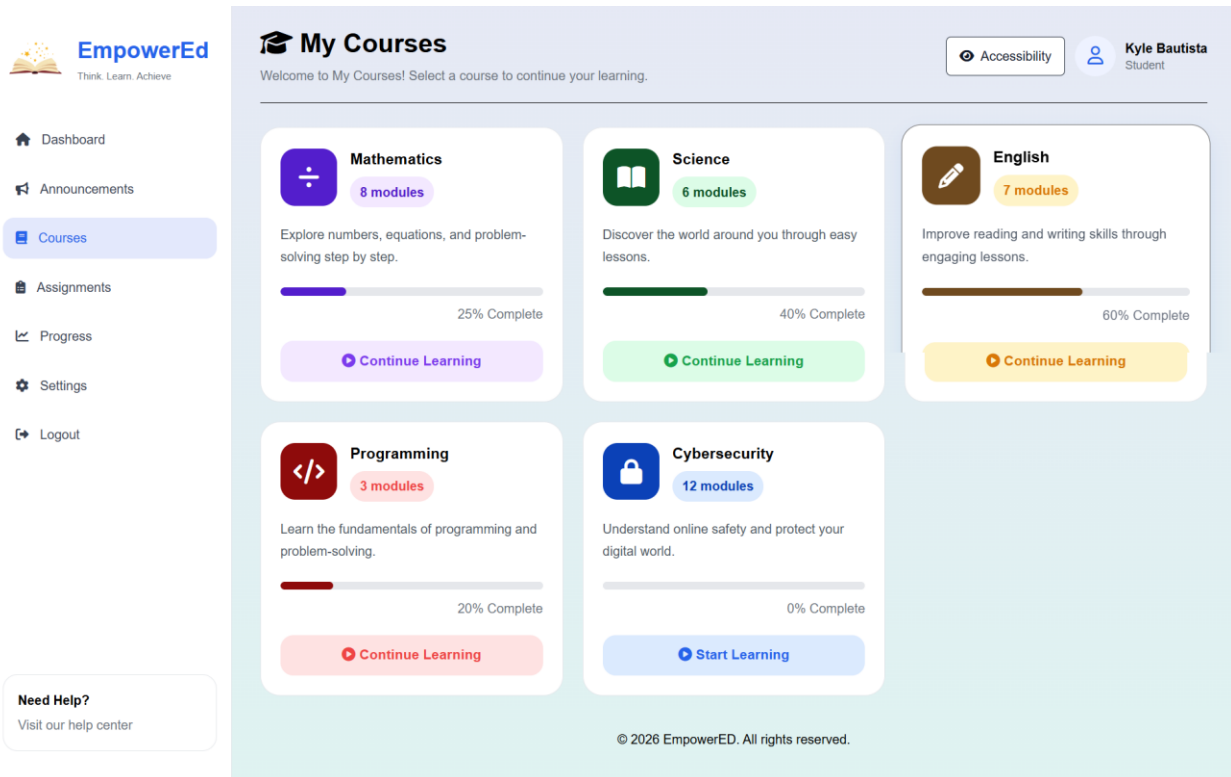
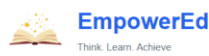


Figure 15: Courses Module

User Interface: Course Module

Figure 15 shows the Course Module where students can view their enrolled courses. The developers added figures and colors for each course to allow easier navigation and improve visual recognition. A progress bar is also included to help students track their learning progress and monitor completed tasks and activities.



- Dashboard
- Announcements
- Courses
- Assignments
- Progress
- Settings
- Logout

Need Help?
Visit our help center

Assignments

Track and manage your assignments.

Accessibility

Kyle Bautista
Student

Upcoming Submitted Missing



Quiz 1
Mathematics

DUE DATE
May 20, 2026

POINTS
10 pts

STATUS
Due Soon

Start >



Quiz 2
Mathematics

DUE DATE
May 28, 2026

POINTS
10 pts

STATUS
Upcoming

Start >



Essay 1
English

DUE DATE
June 2, 2026

POINTS
30 pts

STATUS
Upcoming

View >



Worksheet 1
Mathematics

DUE DATE
June 5, 2024

POINTS
15 pts

STATUS
Upcoming

View >

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Figure 16: Assignments Module

User Interface: Assignments Module

Figure 16 presents the assignments module where students can view all of their assignments and tasks in all courses arranged by date of submission. The developers added figures and colors for each course to allow easier navigation and improve visual recognition. A progress bar is also included to help students track their learning progress and monitor completed tasks and activities.

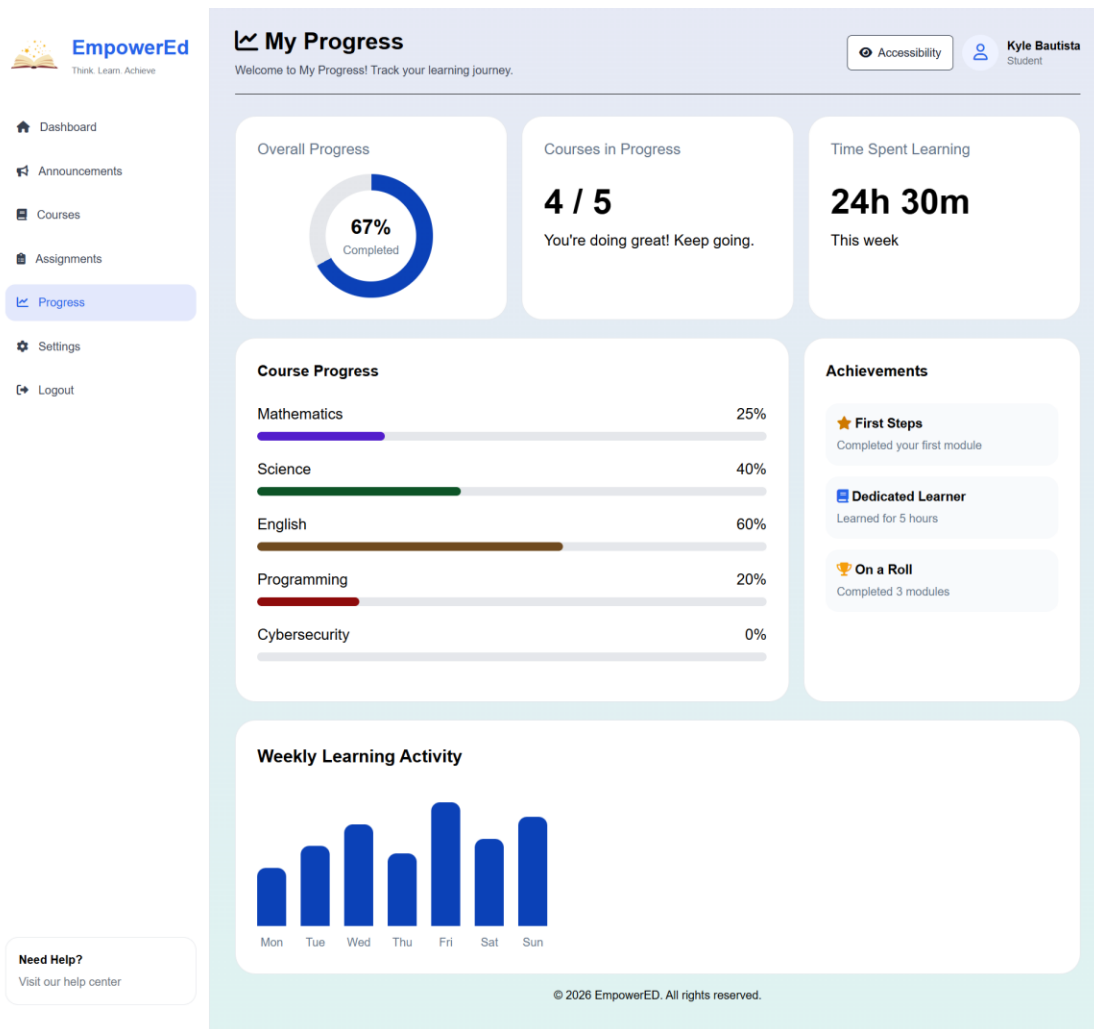


Figure 17: Progress Tracking Module

User Interface: Progress Tracking Module

Figure 17 illustrates the progress tracking module wherein the students can view their overall progress. The developers used charts for data visualization and included achievements and milestone indicators to motivate students and help them, their teachers, and parents to easily monitor their academic performance and completed activities.

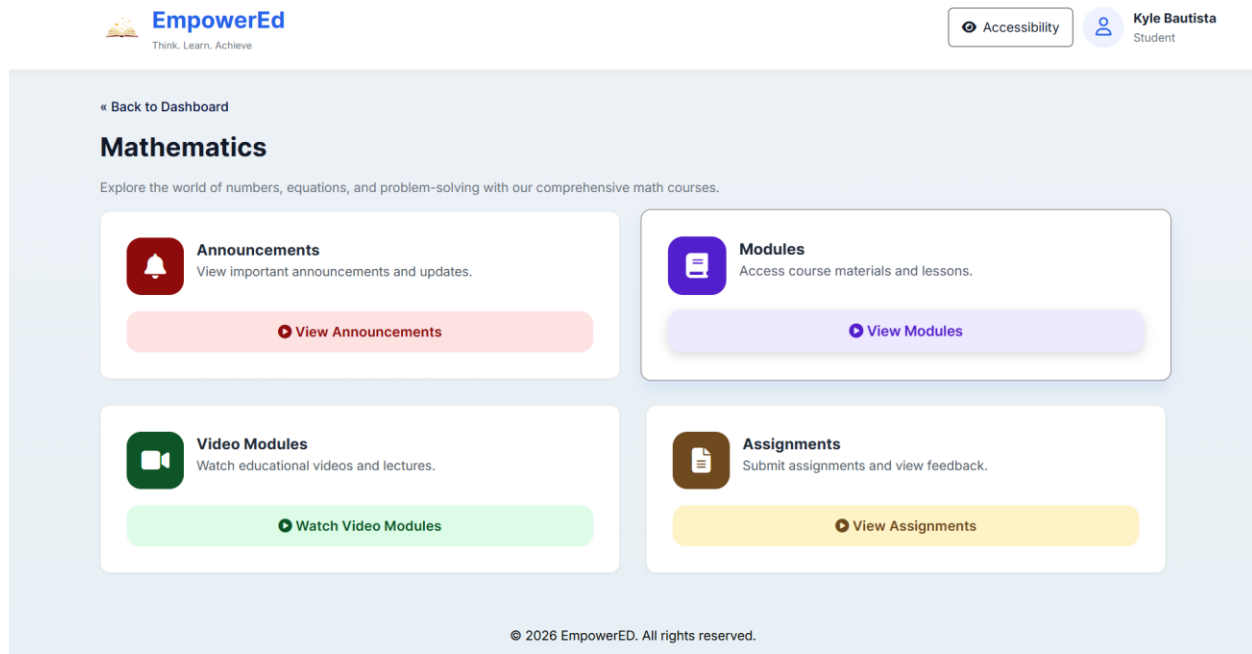


Figure 18: Course Module Buttons

User Interface: Course Module Buttons

Figure 18 shows the contents inside a course. When a student clicks a specific course, the system displays the course announcements, course modules, video modules, and course assignments. Figures and colors were implemented for easier visualization and navigation.



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Course Announcements

Stay updated with important course information and deadlines

🚩 Course Extension: Assignment Deadline Updated

 Read Aloud

 Dr. Joachim Barboza  May 20, 2026 **Important**

Due to technical difficulties, Assignment 3 has been extended to April 27, 2026.

🚩 New Accessibility Resources Available

 Read Aloud

 Dr. Alessandra Metrillo  April 18, 2026

New screen reader tools and accessibility resources are now available in Module 2.

🚩 Quizzes Now Available: Module 2 & 3 Assessment

 Read Aloud

 Dr. Ken Gonzales  April 24, 2026

Module 2 and Module 3 quizzes are now uploaded. Please check Pending Activities.

🚩 Final Project Guidelines Reminder

 Read Aloud

 Dr. Neomi Gabejan  April 25, 2026 **Important**

Please be informed that the Final Project Guidelines have now been released. Kindly take time to carefully read and understand all the requirements, instructions, and deadlines provided to ensure proper compliance.

🚩 Peer Review Activity Deadline Reminder

 Read Aloud

 Dr. Kyle Bautista  April 26, 2026

All students must complete the peer review activity before the deadline. Late submissions will not be accepted.

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Figure 19: Course Announcements

User Interface: Course Announcements

Figure 19 displays the course announcements. It allows the student to view announcement for a specific course. “Read aloud” button was also implemented. “Important” badges were added for easier identification and recognition of essential announcements.

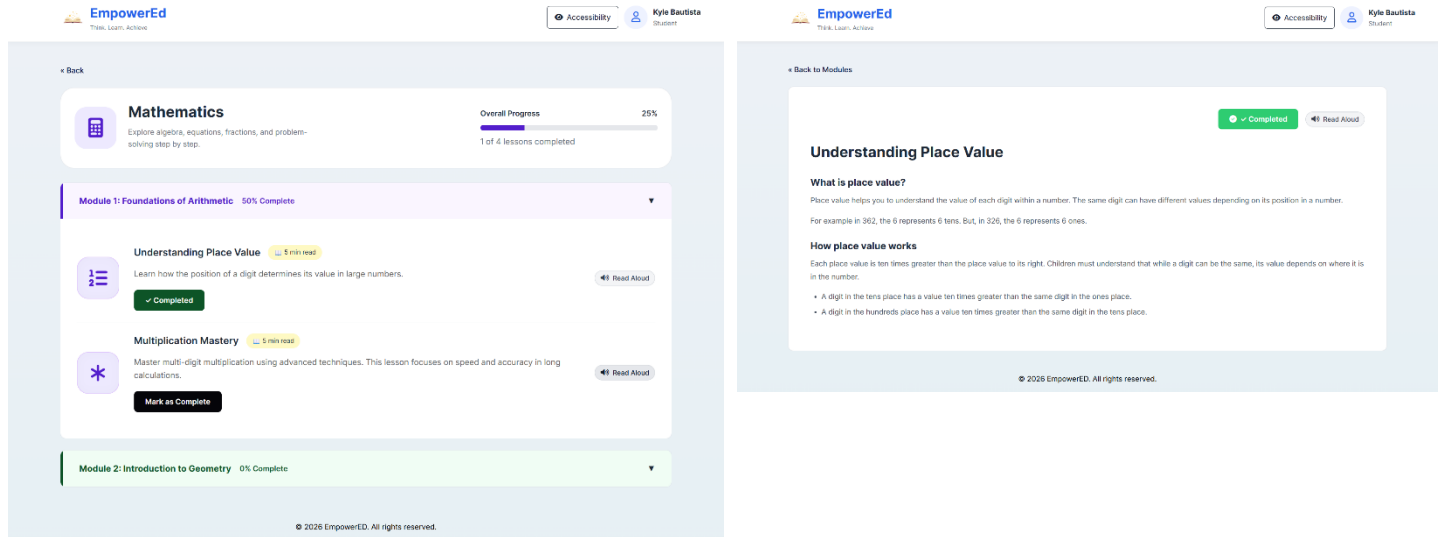


Figure 20: Course Announcements

User Interface: Course Modules

Figure 20 illustrates the Course Modules section, which allows students to view and navigate learning modules for a specific course. A progress bar was integrated to enable easier tracking of learning progress. “Read Aloud” feature was added to enhance accessibility and support students with reading difficulties or visual impairments.



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Accessibility

Kyle Bautista
Student

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Video Modules

Watch recorded lectures and video content for Mathematics.

Grade 4 - Mathematics - Rounding Off / WorksheetCloud Video Lesson

WorksheetCloud

Grade 4
Online Classroom

hosted by
WorksheetCloud

Watch on YouTube

Grade 4 - Mathematics - Rounding Off Video Lesson

In this Grade 4 Mathematics video lesson we will be teaching you about Rounding Off.

29:49

Captions Available Transcript Available Adjustable Speed

Mark as Done

Transcript

[0:00] good morning great forces lovely to have you with me this morning let's jump straight into the lesson without me

[0:06] making a fool of myself by saying some in n things

[0:13] the usual beginning of the lesson just a reminder if you've got any questions great forward worksheet loud calm and

[0:19] off lesson click on the link above the video and do the worksheet just to make sure that you've understood

[0:23] today's list as per normal the next slide is going to be mental arithmetic

Course Playlist

- ▶ Rounding Off Video Lesson
29:49
- ▶ Adding and Subtracting Fractions
36:35
- ▶ 2D Shapes
37:11

Progress 0 of 3 completed

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Figure 21: Course Video Modules

User Interface: Course Modules

Figure 21 illustrates the Video Modules, which allows students to view and watch videos uploaded as learning materials. A progress bar was integrated to enable easier tracking of learning progress. Captions and transcripts were integrated to help people with hearing difficulties.



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Student

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Course Activities

Complete assignments, quizzes, and discussions to earn points and progress in the course

Pending
3
Activities and Quizzes to complete

Completed
1
Activities finished

Total Points
15/55
Points earned

Pending Activities

Module 2: Worksheet
Worksheet for Module 2
Due: June 02, 2026 15 points Upcoming
View Assignment

Quiz: Module 2 Assessment
Mathematics Quiz
Due: May 20, 2026 10 points Due Soon
Start Quiz...

Quiz: Module 3 Assessment
Mathematics Quiz
Due: May 28, 2026 10 points Upcoming
Start Quiz

Completed Activities

Module 1: Worksheet
Submitted: April 13, 2026 Score: 15/20 Completed

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Figure 22: Course Activities

User Interface: Course Assignments

Figure 22 illustrates the Course Activities section, which allows students to view tasks and assignments for a specific course. The developers incorporated progress cards to enable efficient progress tracking. Additionally, icons, badges, and colors were implemented to improve visualization and provide easier navigation and identification of activities.



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Accessibility  **Kyle Bautista**
Student

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Activity Instructions
Answer the following questions and upload your solutions below:

Problem	Answer
What is the value of the underlined digit in 5,682?	
Round 7,349 to the nearest hundred.	
$\$12,450 + \$8,321 = \$$	
$\$5,000 - \$2,674 = \$$	
$\$32 \text{ times } 4 = \$$	
Find the next two numbers in the sequence: 2, 6, 10, 14, ____	
What number is 1,000 less than 10,500?	
A rectangular garden is 10 meters long and 7 meters wide. What is the total distance around the garden?	
A square tile has a side length of 9 inches. What is the area of the tile?	
If a movie starts at 4:15 PM and lasts for 1 hour and 30 minutes, what time does it end?	
A school cafeteria ordered 8 crates of apples. Each crate contains 40 apples. During lunch, students ate 150 apples. How many apples are left over?	

Submission

Comments...

Drag files or [click to browse](#)

Submit

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Figure 23: Activity Panel

User Interface: Activity Panel

Figure 23 displays the Activity Panel, which allows students to complete activities uploaded by their teachers. Students may use the text box to encode their answers or use the upload box to submit their completed works and requirements. The developers designed the interface to be simple and user-friendly to ensure easier access and improved usability for students.

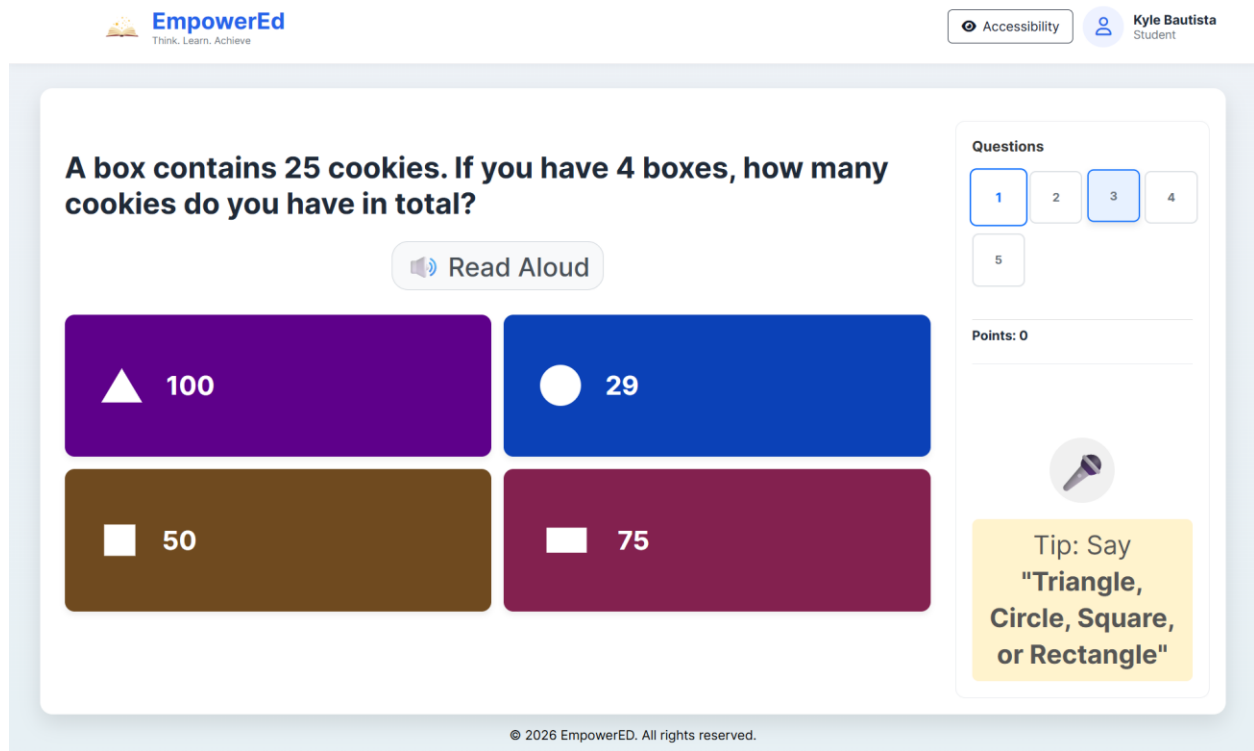


Figure 24: Quiz Panel

User Interface: Quiz Panel

Figure 24 illustrates the Quiz Panel, which allows students to answer quizzes and activities. The developers designed the interface to be simple and intuitive, incorporating larger UI components and text for improved readability. Text-to-Speech and Speech-to-Text features were also implemented to enhance accessibility and support students with diverse learning needs.

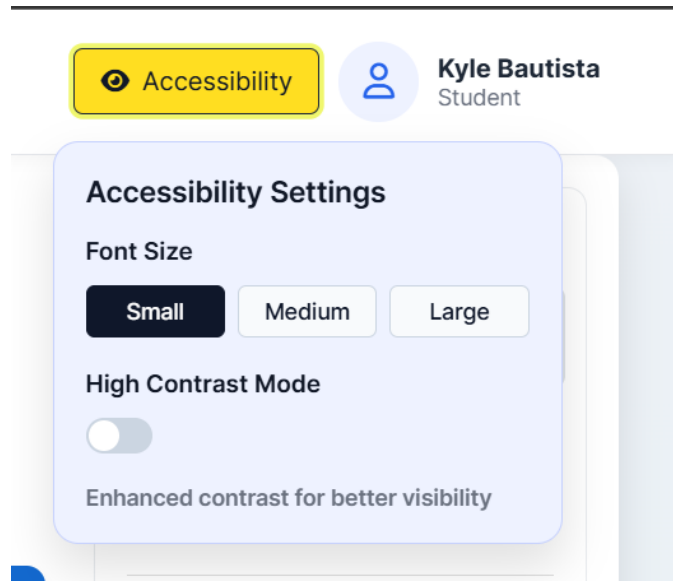


Figure 25: Accessibility Button

User Interface: Accessibility Button

Figure 25 illustrates the Accessibility Button, which allows users to modify the interface display according to their preferences. It provides options for small, medium, and large text sizes, as well as a High Contrast mode to enhance readability and improve accessibility for users with visual impairments.

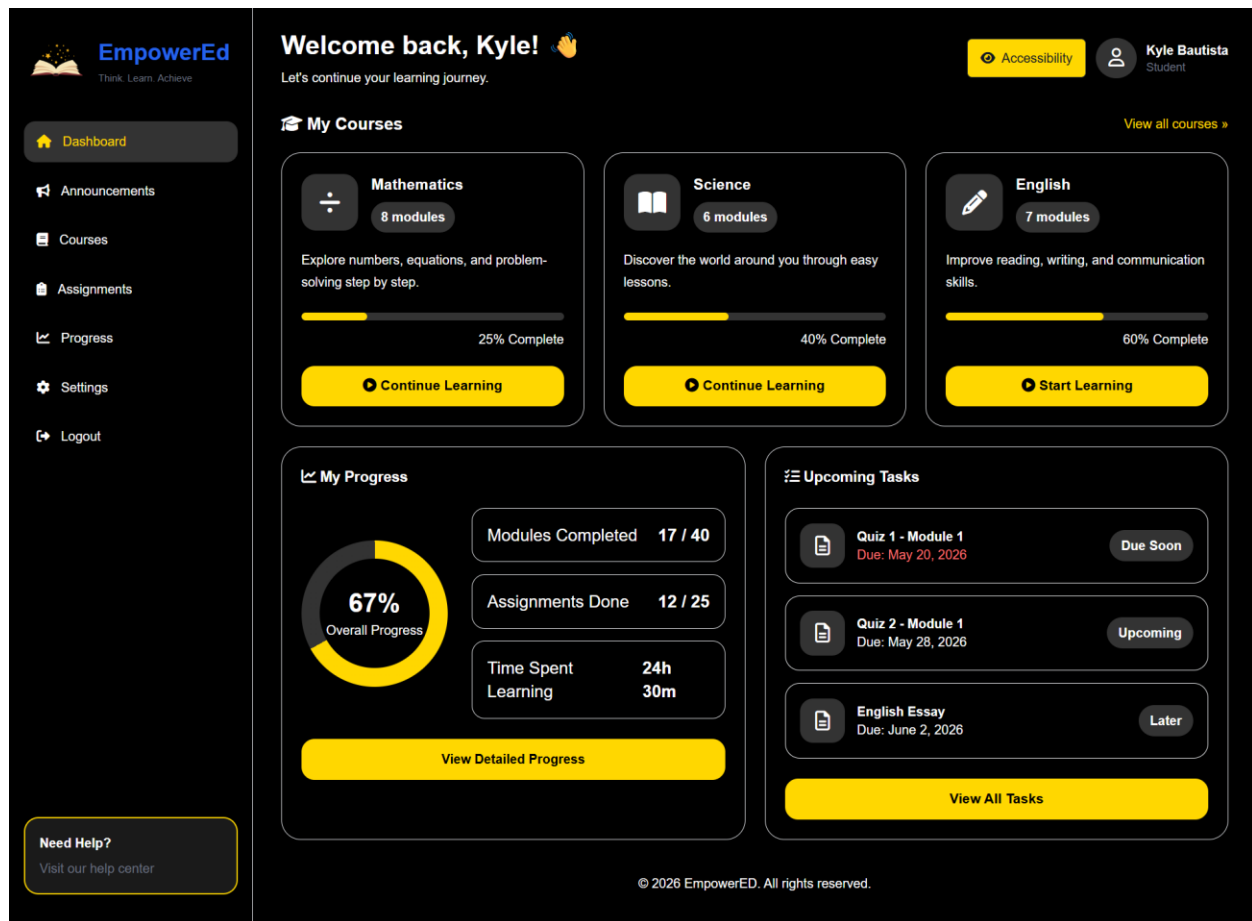


Figure 26: High-Contrast Mode

User Interface: High-Contrast Mode

Figure 26 shows the high-contrast mode of the system. It used yellow, black, and white as primary colors to enhance the visual clarity. This feature is designed to support users with visual impairments and improve overall readability and accessibility.



Development Procedures

The development of the system followed a structured process to ensure that the Learning Management System is functional, accessible, and user-friendly for students with disabilities. The developers utilized modern web technologies, frameworks, and development tools throughout the project.

Programming Languages Used

EmpowerEd: Think. Learn. Achieve. was developed by the developers who created and designed a web-based Learning Management System (LMS) using modern web development techniques. In the front-end development, the system uses HTML5, CSS3, and JavaScript, creating an interactive interface for its users. As for the back-end development, the system uses Node.js and Express.js to perform server-side operations and application management. And for the system, it uses MongoDB, a NoSQL solution for database management.

Framework and Tools

To enhance the system's visual appeal and improve the user interface experience, several tools and frameworks were utilized in the project. These include Font Awesome for icons, Google Fonts for typography, CSS Flexbox for layout design, and Bootstrap for responsive design features. Moreover, developers tested the colors used in the system and ensures that it passed the Web Content Accessibility Guidelines by using WCAG Color Contrast Checker. Lastly, Visual Studio Code was used as the coding editor, while GitHub served as the platform for collaborative project development.

Coding Procedures

The programmers designed the system to be accessible and easy to use by programming features and design principles that are inclusive to students with disabilities, thus making the program a more flexible learning environment. They utilized modular and accessibility-oriented approaches during the implementation. First, they developed the front-end components needed for navigation through the interface and processes. Next, they included and added the accessibility features, which include high-contrast mode, resizable fonts, speech-to-text conversion, text-to-speech conversion, transcripts, and captions. And lastly, they incorporated the database and then proceeded with implementing back-end functionalities using RESTful APIs. Regular debugging and testing procedures were also conducted to ensure system reliability and performance.

Module Development

Modules were created to organize the system better. This helps in the testing and debugging process. These modules include User Authentication, Dashboard, Announcements, Courses, Quiz, Assessments, Video Modules, Modules, Progress Tracking, Accessibility Features, Settings, and Profile. Each module was developed, tested, and then integrated into the system to ensure proper functionality.

Testing and Evaluation Procedures

To make sure that all integrated components work properly as a cohesive software product, the EmpowerEd platform will go through accurate System Testing. This phase will focus on confirming that the accessible features such as text-to-speech, adjustable font sizes, and subtitle synchronization work seamlessly across various modules and satisfy the requirements for students with disabilities. This stage focuses on confirming that the accessible features like text-to-speech, font size adjustments, and subtitle synchronization meet the needs for students with disabilities and function flawlessly across various courses.

The evaluation of the system will be conducted based on the ISO/IEC 25010 Software Quality Standards using the following criteria:

1. **Functionality:** Assessing whether the platform provides all necessary tools, such as lesson uploads and progress monitoring, to meet the educational needs of users.
2. **Usability:** Evaluating the ease with which students with disabilities can navigate the interface and utilize accessibility features effectively.
3. **Reliability:** Testing the platform's ability to maintain consistent performance and accessibility under various conditions.
4. **Efficiency:** Measuring the system's responsiveness and resource usage during active learning sessions.
5. **Security:** Ensuring that student progress data and teacher materials are protected from unauthorized access.



Statistical Treatment of Data

To analyze the feedback gathered from users regarding the platform's effectiveness and accessibility, the primary statistical tool used will be Likert Scale Interpretation. Survey responses will be collected using a 5-point scale (from Strongly Agree to Strongly Disagree) to quantify user satisfaction and the perceived impact on reducing learning barriers.

Data gathered through this method will allow the research team to:

1. Calculate the Weighted Mean of responses for each evaluation criterion.
2. Evaluate the collective agreement between teachers and learners concerning how well the platform promotes inclusive education.
3. Identify specific areas of the platform that require further technical refinement based on numerical trends.

References

- Abonitalla, J. R., & Dejarme, J. J. (2025). Level of use of technology learning tools and learning outcomes among junior and senior high school learners. Philippine E-Journals. <https://ejournals.ph/article.php?id=32007>
- Ahmed, I. (2024). THE INTERSECTION OF DISABILITY AND HUMAN RIGHTS: EXPLORING THE SOCIAL, LEGAL, AND ETHICAL DIMENSIONS OF DISABILITY RIGHTS IN CONTEMPORARY SOCIETIES. (2024). *Frontiers in Multidisciplinary Studies*, 1(01), 21-30. <https://fmsjournal.com/index.php/journal/article/view/3>
- Alfonso, G. J., & Garcia, P. G. (2021). *Open and Distance e-Learning (ODEL): Shaping the Future of Teaching and Learning*. University of the Philippines Open University. <https://networks.upou.edu.ph/wp-content/uploads/2021/08/ODEL-shaping-the-future-of-teaching-and-learning-SINGLE-PAGE-wth-cover.pdf>
- Altınay, F., & Altınay, Z. (Eds.). (2024). Intellectual and learning disabilities: Inclusiveness and contemporary teaching environments. IntechOpen. <https://www.intechopen.com/books/12454>
- Arantes, J. (2025). Reframing AI governance in education: Insights from the social model of disability. *Learning, Media and Technology*, 1–16. <https://doi.org/10.1080/17439884.2025.2595443>



Assistive Technology Industry Association (ATIA). (2024). *Assistive technology outcomes and benefits (ATOBS)* (Vol. 18). https://www.atia.org/wp-content/uploads/2024/04/ATOBS_V18_FINAL-1.pdf

Asthana, P., & Gupta, V. K. (2025). Using automatic speech recognition to support students with disabilities. *UKLO PROCEEDINGS*, 1(1), 187-193. <https://doi.org/10.20544/AISC.1.1.25.P18>

Baidya, A., & Baidya, M. (2025). Teachers' Preparation For Imparting Education To PWD Students In Secondary Schools - A Comprehensive Review. *IOSR Journal Of Humanities and Social Science*, 30, 12-18. <https://www.bing.com/ck/a?!&&p=8b410590ad666e55ab4fef881b0a8ceaba60015b2afbd9b75982198e7ea548c9JmItdHM9MTc3OTQwODAwMA&pptn=3&ver=2&hsh=4&fclid=250e0f34-45c6-64ac-0305-1932445b65d1&psq=Teachers%e2%80%99+Preparation+For+Imparting+Education+To+PWD+Students+In+Secondary+Schools+-+A+Comprehensive+Review&u=a1aHR0cHM6Ly93d3cuaW9zcmpvdXJuYWxzLm9yZy9pb3NyLWpoc3MvcGFwZXJzL1ZvbC4zMC1Jc3N1ZTUvU2VyLTcvQzMwMDUwNzEyMTgucGRm>

Blicher et.al. (2025). *Doing the Work: Diversity, Equity, and Inclusion in Open Educational Resources*. Pressbooks. <https://openoregon.pressbooks.pub/dothework/chapter/2-1-overview/>



- Chao, C. M., et al. (2022). Digital technology and increasing engagement among students with intellectual disabilities. *Procedia Computer Science*, 214, 112-119.
<https://www.sciencedirect.com/science/article/pii/S2666557322000283>
- Chavan, R. R., Patil, S. D., Patil, I. R., Patil, S. K., & Yadav, P. M. (2025). EduLife: Learning Platform for Specially-Abled Students.
<https://doi.org/10.36227/techrxiv.176289238.87628954/v1>
- Digital Policy Office. (2024). *Handbook on web and digital accessibility*. The Government of the Hong Kong Special Administrative Region.
https://www.digitalpolicy.gov.hk/en/our_work/digital_government/digital_inclusion/accessibility/promulgating_resources/handbook/
- Garcia, M. B. (2025). Evaluating the usability of Canvas LMS on PWA and native mobile platforms: A role-based comparison of student and teacher experience. Manuel B. Garcia Research Publications / Philippine Context.
https://manuelgarcia.info/media/full_paper/usability-canvas-lms.pdf
- Guidelines on the Inclusion of Learners with Disabilities in Open and Distance Learning*. (2025, September 18). <https://www.unesco.org/en/communication-information/odl-guidelines>
- Gupta, S., & Chen, J. (Eds.). (2025). *Personalized Learning Environments: Core Concepts, Technologies, and Practices*. Springer.
<https://doi.org/10.12937/itel.4.1.inv.p002>



- Hornby, G., & Kauffman, J. M. (Eds.). (2021). *Special and inclusive education: Perspectives, challenges and prospects*. MDPI. [https://mdpi-res.com/bookfiles/book/4506/Special and Inclusive Education Perspectives Challenges and Prospects.pdf](https://mdpi-res.com/bookfiles/book/4506/Special_and_Inclusive_Education_Perspectives_Challenges_and_Prospects.pdf)
- Ilocos Sur Polytechnic State College. (2021). *E-Dawa institutional repository*. <https://ispsc.edu.ph/e-dawa-tbgn2157>
- Inclusion in education*. (2026, January 30). UNESCO. <https://www.unesco.org/en/inclusion-education>
- Kasimakhunova, A., & Atajonova, S. (2025). *Concept, principles and evolution of inclusive education*. *American Journal of Social Sciences and Humanity Research*, 5(5), 331–335. <https://doi.org/10.37547/ajsshr/Volume05Issue05-66>
- Langcuyan, V. R., Lopez, I. C., & Mendez, M. L. S. P. (2024). Enhancing reading comprehension in Araling Panlipunan of Grade 5 learners through differentiated instruction. *Davao Research Journal*, 15(4), 95-107. <https://ejournals.ph/article.php?id=21404>
- Lomellini, A., Lowenthal, P. R., Snelson, C., & Trespalacios, J. H. (2025). Accessible and inclusive online learning in higher education: a review of the literature. *Journal of Computing in Higher Education*, 37(3), 1306–1329. <https://link.springer.com/article/10.1007/s12528-024-09424-2>
- Lynch, P., Singal, N., & Francis, G. A. (2024). Educational technology for learners with disabilities in primary school settings in low- and middle-income countries: a



systematic literature review. *Educational Review*, 76(2), 405–431.

<https://doi.org/10.1080/00131911.2022.2035685>

Mastam, N. M., & Zaharudin, R. (2024). Work Readiness Skills for Students with Learning Disabilities in Special Education Vocational Schools: A Conceptual Framework.

Jurnal Pendidikan Bitara UPSI, 17, 1–10.

<https://doi.org/10.37134/bitara.vol17.sp.1.2024>

Mehmood, M. U., & Parveen, Z. (2025). EMPOWERING TEACHERS FOR INCLUSIVE EDUCATION: A SUPPORT PROGRAM FOR CHILDREN WITH DISABILITIES IN PUNJAB'S ELEMENTARY SCHOOLS. (2025). *International Journal of Social Sciences Bulletin*, 3(3), 390-405.

<https://www.researchgate.net/publication/3898889856> EMPOWERING TEACHERS FOR INCLUSIVE EDUCATION A SUPPORT PROGRAM FOR CHILDREN WITH DISABILITIES IN PUNJAB'S ELEMENTARY SCHOOLS

Nurali, Q. D. (2025). PRINCIPLES OF INCLUSIVE EDUCATION AND ITS NEEDS. American Academic Publishers.

<https://www.academicpublishers.org/journals/index.php/ijai/article/view/2460/356>

2

Pagtaconan, W. C. R., & Mercado, C. A. (2021). *Design of a virtual learning environment implementation framework for state universities in the Philippines: A case of Mariano Marcos State University*. *International Journal on Open and Distance e-Learning*. <https://share.google/ILILM3ixJ7FdcXfbT>



- Pavani, F., & Leonetti, V. (2025). On the coexistence of captions and sign language as accessibility solutions in educational settings. *Sensors*, 26(3), 852. <https://doi.org/10.3390/s26030852>
- Pegalajar-Palomino, M. C. (2022). Implications of mobile learning for sustainable inclusive education: A systematic review. *The Electronic Journal of e-Learning*, 20(5), 538–553. <https://www.ejel.org>
- Priyadharsini, V., & Sahaya Mary, R. (2024). Universal Design for Learning (UDL) in inclusive education: Accelerating learning for all. *Shanlax International Journal of Arts, Science and Humanities*, 11(4), 145-150. <https://doi.org/10.34293/sijash.v11i4.7489>
- Republic Act No. 10754. (2016). *An Act Expanding the Benefits and Privileges of Persons with Disability (PWD)*. Official Gazette of the Republic of the Philippines. <https://www.officialgazette.gov.ph/2016/03/23/republic-act-no-10754/>
- Rogers, J. K. B., Salazar, R. P., & Buladaco, M. V. M. (2025). *Moodle and Google Classroom: A comparative study of acceptability*. *Journal of Education and Learning*, 19(3), 1670–1677. <https://zenodo.org/records/17973463>
- Setiawan, A. (2024). Intelligent systems for disabilities: Technology for a more inclusive life. SSRN. <https://ssrn.com/abstract=5062273>
- Sherafim, Z. (2026). Development of interactive multimedia based on gamification learning to improve learning outcomes for children with disabilities. ALACRITY:



Journal of Education, 6(2).

<https://lppipublishing.com/index.php/alacrity/article/view/1003>

Slootman, M., Korthals Altes, T., Nielandt, B., Reynders, K., Domagała-Zyśk, E., Rodríguez-Ardura, I., & Stanojev, I. (2023). A handbook of e-inclusion: Building capacity for inclusive higher education in digital environments. Knowledge Innovation Centre. ISBN: 978-99957-79-09-2. https://einclusion.net/wp-content/uploads/sites/43/2023/02/E-Inclusion_Handbook-04-02-2023-12_02_14_617.pdf

Takacs, S., Zhang, J., Lee, H., Truong, L., & Smulders, D. (2022). *A comprehensive guide to applying Universal Design for Learning*. BCcampus. <https://pressbooks.bccampus.ca/jibcudl/open/download?type=pdf>

Teacher Task Force. (2020). *Assistive technologies: Inclusive teaching guidelines for educators*. UNESCO. https://teachertaskforce.org/sites/default/files/2024-09/2020_KS_AssistiveTech_Guidelines.pdf

U.S Department of Education. (2022). *Alternative Learning System (ALS) strategic plan* (Version 6). Republic of the Philippines. <https://www.deped.gov.ph/wp-content/uploads/2022/06/ALS-Strategic-Plan-V6-Final-Web.pdf>

U.S. Department of Education, Office of Educational Technology. (2023). *Artificial intelligence and the future of teaching and learning: Insights and recommendations*. Washington, DC. <https://www.ed.gov/sites/ed/files/documents/ai-report/ai-report.pdf>



U.S. Department of Education, Office of Educational Technology. (2023). *Artificial intelligence and the future of teaching and learning: Insights and recommendations*. Washington, D.C.
<https://www.ed.gov/sites/ed/files/documents/ai-report/ai-report.pdf>

UN General Assembly. (2006). *Convention on the Rights of Persons with Disabilities: resolution / adopted by the General Assembly*. December 13, 2006, A/RES/61/106.
<https://www.refworld.org/docid/45f973632.html>

UNESCO. (2021). *Reaching out to all learners: A resource pack for supporting inclusive education*. UNESCO International Bureau of Education.
https://www.ibe.unesco.org/sites/default/files/medias/fichiers/2023/10/unesco_bie_2021_web_inclusive_education_resrouce_pack.pdf

United Nations. (2026). *Goal 4: Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all*. Sustainable Development Goals.
<https://sdgs.un.org/goals/goal4>

Waisath, W., McCormack, M., Stek, P., & Heymann, J. (2024). Dismantling barriers and advancing disability-inclusive education: an examination of national laws and policies across 193 countries. *International Journal of Inclusive Education*, 28(10), 2088–2103. <https://doi.org/10.1080/13603116.2022.2058623>

Yang, M., Duha, M. S. U., & Luo, T. (2024). Universal design in online education: A systematic review. *Distance Education*.



https://www.semanticscholar.org/paper/Universal-design-in-online_education%3AA-systematic-Yang-Duha/5d416f2c346a5e0d311da22cd81bbe41ec2600ba

Appendix A

Gantt Chart



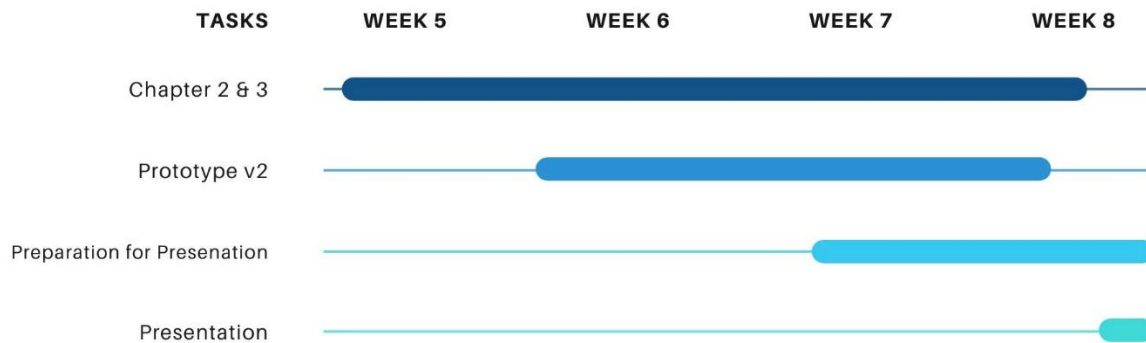
EmpowerED Gantt Chart

Start: April 7 | End: May 29



EmpowerED Gantt Chart

Start: April 7 | End: May 29





Appendix B

High-Contrast Mode

Appendix C**WCAG Test Result**



**NATIONAL
UNIVERSITY**

NATIONAL UNIVERSITY, MANILA
SY. 2025-2026



NATIONAL UNIVERSITY
COLLEGE OF COMPUTING AND
INFORMATION TECHNOLOGIES